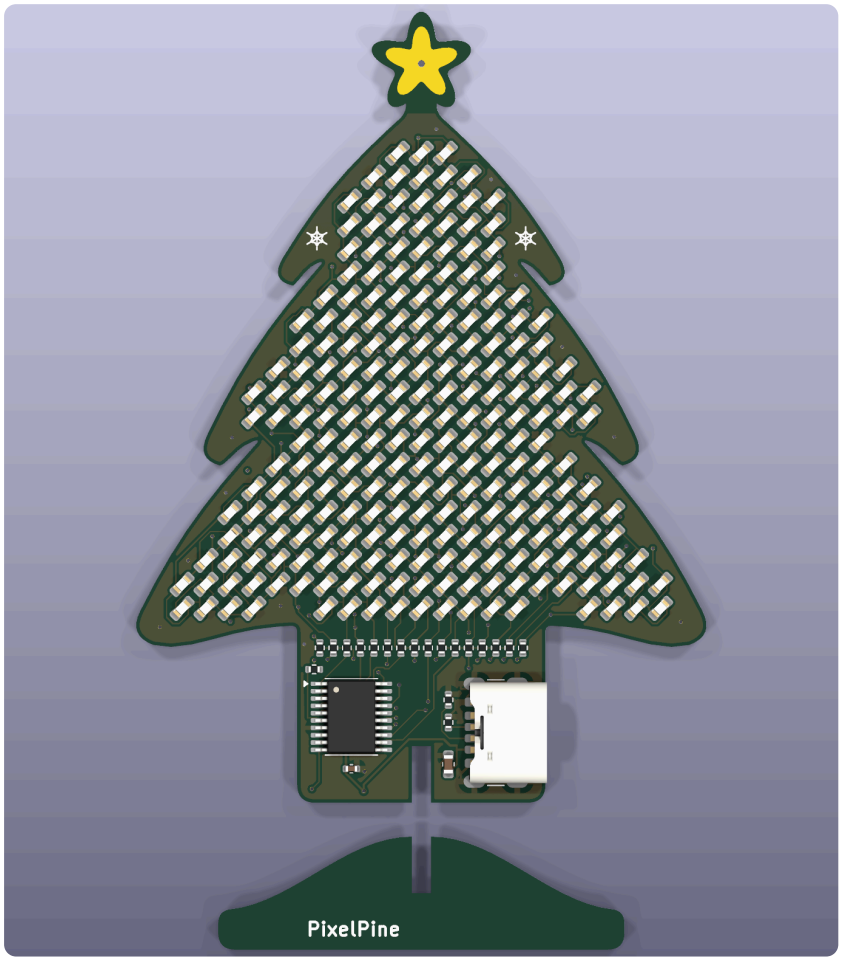
PixelPine

A 7 cm PCB Christmas tree with 240 LEDs and a personality. Project documentation, rev A



KTE Project 1 · MachStuff s.r.o. · 2026-09-24

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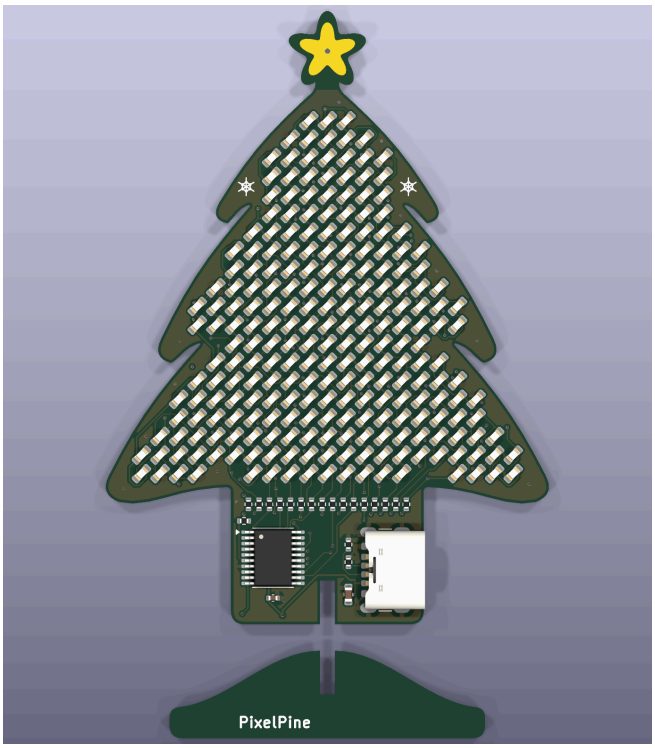
The schematic is a separate PDF: hardware/pixelpine_schematic.pdf.

PixelPine — a 7 cm PCB Christmas tree with 240 charlieplexed LEDs

KTE Project 1 · design-for-manufacture study · rev A · 2026-09-23

A chubby, cute, 70 mm tall PCB tree covered in **240 warm-white 0603 LEDs** on a 2.12 mm grid. The LEDs are charlieplexed on 16 pins of a **$0.13 Puya PY32F003** (Cortex-M0+, 64 KB flash) with software PWM. It runs from USB-C and slots into a PCB stand fin. The tree is a character, **Pine**, whose job is showing light shows. He's bad at it: he daydreams, naps, gets grumpy and gets excited to show you things. You interact with him through the capacitive star, which senses tap, pet, press, a front-and-back squeeze and tickles. The product firmware runs on the first boards and matches the simulator frame for frame (`firmware/pixelpine/`). Landed cost ≈ **€3.2 at 1 000 pcs, €2.0 at 10 000 pcs**; suggested shelf price **€12.90**.

Try Pine in your browser: machmar.github.io/pixelpine (the animation simulator).

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Deliverables

You asked for	Where
Schematic	`hardware/pixelpine_schematic.pdf` · KiCad 10 project `hardware/pixelpine.kicad_pro` (ERC: 0 errors, 0 warnings)
PCB shape and layout	`hardware/pixelpine.kicad_pcb`: outline, stand fin, all 267 parts (+2 logo marks) placed with nets, silkscreen, CE/WEEE marks, **routed on 2 layers** (Freerouting, then finished by hand in KiCad: GND pour on both layers, 142 vias). DRC: 0 unconnected, schematic parity 0 issues; 2 cosmetic *starved thermal* notes (see the order guide). The LEDs sit in a routing-friendly arrangement: see LED arrangement
BOM (LCSC / JLCPCB)	`docs/bom.md` · JLC upload files `hardware/production/jlc_bom.csv`, `jlc_cpl.csv`
Order files	`hardware/production/`: Gerber + drill zips for the tree (PCB + assembly) and the stand fin (bare PCB), BOM/CPL, 3D previews, and an order guide with every JLC setting and what to check in the placement preview

You asked for	Where
PC app to preview the animations	`simulator/index.html`, live at machmar.github.io/pixelpine (published from `simulator/` by `.github/workflows/pages.yml`): open it in any browser; works offline apart from fonts. Pine runs live: touch his star (hover / click / hold / shift / ctrl / fast clicks) and watch his feelings change. Preview mode plays any show, face or transition. Includes an electrical model of the charlieplex (current sag, PWM depth, refresh, pin current)
Pine: character, faces, transitions, gestures	`docs/animations.md`: behaviour state machine, emotions, 15 light shows, 16 faces, 6 transitions, touch gestures, RAM/flash budget
Market analysis	`docs/market-analysis.md`
EU certification setup	`docs/eu-compliance/`: applicability matrix, EMC test plan, technical file + risk assessment, DoC template, labels and leaflet (EN/CZ)
Final price analysis incl. assembly	`docs/cost-analysis.md`, generated by `docs/cost_model.py`
Connectivity & power consumption	`docs/power-and-connectivity.md`
Product firmware (beyond the brief)	`firmware/pixelpine/`: the simulator ported to C and running on the board: **59 KB of 63 KB flash, 6.9 KB of 8 KB RAM**, 50 fps, 122 Hz BCM scan. Checked frame by frame against the simulator's own JavaScript: Pine's state machine and emotions identical in every whole-Pine test, LED levels within 1 step of 255 on ≥ 99.99 % of LED-frames there (per show: 93–100 %, fireflies lowest). Touch calibrated on the board; room-light sensing isn't possible on rev A (and how rev B fixes it); a whole-tree touch field was tried and dropped (why). Flashing: `firmware/flash/`
Firmware fit check	`firmware/sizetest/`: real compile of all shows, faces, brain and drivers for both MCUs. The original CH32V003 overflowed (20.9 KB of 16 KB), so the MCU was switched to the PY32F003F18P6: **17.2 KB of 64 KB flash (27 %), 1.9 KB of 8 KB RAM (23 %)**, built with the Arm GNU Toolchain. See `docs/animations.md`
Design spec	`docs/superpowers/specs/2026-09-23-pixelpine-design.md`
All docs as PDF	`docs/pdf/PixelPine-documentation.pdf`: everything above in one linked PDF (cover, contents), plus one PDF per doc in `docs/pdf/`. Regenerate with `python docs/export_pdfs.py` after editing a doc

Key numbers

LEDs	240 × XL-1608WWC-06 (0603, 2500–3000 K) at 45°, 2.12 mm pitch
MCU	Puya PY32F003F18P6 (TSSOP-20, pinout 1, 24 MHz internal oscillator): 16 charlieplex lines (incl. SWD pins and BOOT0) + ADC touch star (PA7) + NRST; SWD programming
Character	6 emotions (energy, focus, mood, annoyance, excitement, bond in flash), 16 faces, 6 transitions
Parts	263 placements, 8 line items, 552 SMD joints, single-sided
PCB	2-layer FR-4 1.6 mm, LF-HASL, JLC standard process (0.10 mm tracks and spaces at the narrowest, 0.3 / 0.4 mm vias); 1658 track segments (1991 mm), 142 vias, GND pour on both layers, 0 unrouted. Tree 50.5 × 70 mm, stand fin 36 × 10 mm
Supply	USB-C 5 V, 3–12 mA typical (≈ 7 mA average), 19 mA max → ≤ 0.1 W
Light	≈ 2 lm total, so outside the EU light-source ecodesign/energy-label scope
Pin current	≤ 16.2 mA per GPIO by design (20 mA rated)
PWM	8-bit binary-code modulation, 16 phases, 122 Hz refresh
CE route	EMC 2014/30/EU (EN IEC 55015 + EN 61547) + RoHS; GPSR, WEEE, PPWR, REACH obligations

LED arrangement

The 240 LED positions are a fixed 2.12 mm grid, but *which* anode/cathode pair sits at which position is a free choice, and it decides whether the board can be routed at all. The first version interleaved the pairs (`anode = n mod 16`), which

leaves every one of the 480 LED pads as an island of its own, and Freerouting stalled with 76 connections still open after 87 minutes.

The 45° LEDs form a "brick" lattice with only ~0.55 mm between neighbouring pads. If two neighbouring pads are on the same line net, a tiny F.Cu bridge joins them for free, so a net can be a few big copper islands instead of 30 loose pads. `hardware/scripts/optimize_leds.py` (simulated annealing, C core in `optimize_leds.c`) arranges the 240 pairs for exactly that: 480 separate pads became 141 islands, only the links between islands need real wiring, and the board routes on **2 layers** in about 2 minutes (it also stopped needing the 4 layers that the old arrangement was thought to need). The result is `hardware/led_assignment.json`, committed; `geometry.py` reads it, so the schematic, PCB, simulator map and firmware tables all follow from it. Every LED keeps the same orientation; the optimiser can also turn LEDs round by 180° (`--flip`) for ~15 % fewer islands, but that wasn't needed.

Side effect: neighbouring LEDs now sometimes share an anode line (17 % of the LED pairs closer than 3.2 mm, against 6 % with the old interleaving), so a lit blob loads some scan phases harder than before. The numbers are in `docs/power-and-connectivity.md`.

Regenerating the hardware

> **Since 2026-10-01 the board is finished by hand in KiCad** (GND pour, extra vias). `build_all.py` would regenerate it from the scripts and drop those edits, so it now stops unless you pass `--regenerate`. For order files run `export_gerbers.py` (KiCad's Python) and `export_fab.py`; see `hardware/production/README.md`.

Everything in `hardware/` was generated from one geometry module, so it stays consistent:

```
python hardware/scripts/build_all.py
```

`geometry.py` (outline, LED grid, charlieplex map from `led_assignment.json`) → `gen_footprints.py` → `gen_sch.py` (+ ERC, netlist check) → `gen_pcb.py` (KiCad's bundled Python) → `route.py` (Freerouting, ~2-5 min; if connections stay open it routes again from scratch, up to 4 runs) → DRC and parity → 3D renders → JLC BOM/CPL. It also refreshes `simulator/led_map.js`, so the simulator always shows the real LED positions. After editing the outline, run `python hardware/scripts/geometry.py --search` and paste the new pitch into `GRID_PITCH`, then `python hardware/scripts/optimize_leds.py` (needs a C compiler, 10–20 min) to re-arrange the LEDs for the new grid. The firmware's per-LED tables come from the same map: `python firmware/pixelpine/gen_tables.py`, then `python firmware/pixelpine/build.py flash` builds and programs a board, and `python firmware/pixelpine/test/compare.py` checks the firmware against the simulator.

Open items before ordering

Order files are ready: `hardware/production/README.md`. At order time, check LCSC stock for the 240 LEDs per board and look at the LED polarity and U1 pin 1 in JLC's placement preview.

1. **Firmware: done on the first board** (2026-11-03): it powers up at 4.97 V with no short, flashes through a J-Link (`firmware/flash/`) and runs Pine (`firmware/pixelpine/`). Still open: touch thresholds measured on one board and one hand (measure a batch); the EMC worst-case and production-test firmware builds.
2. **Room light can't be sensed on rev A**: in a full charlieplex every LED has an anti-parallel twin, so no LED can work as a photodiode, and Pine stays in "lights on" mode. Rev B fix at no cost: leave one LED unpopulated (`firmware/pixelpine/README.md`).
3. **Hover isn't detectable** with the star electrode (a finger 1 cm away is below the noise); tap, pet, press, squeeze and tickle work.
4. **No whole-tree touch on rev A**: the LED lines can pick up the mains hum a finger carries, but only on a mains-tied supply (strong on a laptop, weak on a wall charger, nothing on a powerbank), so it was dropped after five experiments (`firmware/pixelpine/README.md`). The star is the touch control.

Done: PCB routed on 2 layers (LED arrangement optimised for it, DRC clean), manufacturer name, address (Jehličková 240, 240 24 Stromečkov: made up for the school project, replace it with the registered address for a real product) and

e-mail on the product, U1 and J1 placement orientation checked, MCU choice final (PY32F003F18P6, kept over the cheaper F002A for firmware-update headroom).

Market analysis — PixelPine

One-line positioning

A **ready-built, cute 7 cm PCB Christmas tree with 240 warm-white LEDs and a personality**: Pine, who is supposed to show you light shows but daydreams, naps, sulks and gets excited instead. Sold as an **affordable-premium gift gadget (€10–15)**. It is not a soldering kit and not a garish RGB flasher. It's a tiny desk companion.

Product category and price tier

Dimension	Where PixelPine sits
Category	Seasonal decorative electronics / desk gadget / small gift (the "Secret Santa" shelf)
Price tier	**Affordable premium**: above €2–5 AliExpress kits, well below €25–35 boutique PCB art
Recommended shelf price	**€12.90** retail incl. VAT · **€14.90** own e-shop incl. VAT · **€6.90 ex-VAT** B2B promo with custom logo (500+ pcs)
Purchase type	Impulse / gift, decided in seconds from a photo or short video. The animation *is* the ad.
Seasonality	~80 % of sales mid-November to 23 December; B2B orders placed September–October

Why this tier works: the landed cost at volume is **€2.0–3.2** (see cost-analysis.md). That leaves room for a retailer's margin at €12.90 once volume passes ~3 000 pcs. In the own e-shop and B2B channels it pays from the first 1 000 pcs.

Target customers

Segment	Who	Why they buy	Channel	Price sensitivity
Gift buyers (primary)	Office Secret Santa, colleagues, teachers, "something small for everyone"	Looks special, has a character people talk about ("mine was grumpy all morning"), needs no batteries	Gift shops, museum/design shops, e-shops, Christmas markets	Medium, with a €10–20 budget ceiling
Tech enthusiasts & makers	Engineers, students, electronics hobbyists	Dense visible SMD matrix, charlieplexing, open animation simulator, SWD pads to hack their own firmware	Tindie, Etsy, own e-shop, maker fairs, r/electronics & Hackaday style posts	Low. They pay for the story and the hackability.
B2B promo	Electronics distributors, engineering firms, universities (faculty gifts), IT companies	A useful, on-brand, non-plastic-junk giveaway. **Logo on the stand fin or back silkscreen costs €0 extra**	Direct sales, promo agencies	High, but orders are 500–5 000 pcs
Kids' rooms (secondary)	Parents	Night-light look	—	Not targeted: the product is *not a toy* (small parts, exposed electronics) and is labelled so

Competitive landscape

Competitor type	Example	Price seen	What they offer	Our edge
DIY soldering kits (AliExpress)	"3D Christmas Tree soldering practice kit", 3/7-colour flashing LEDs (example, example)	**$2.53–4.33** incl. shipping	12–36 THT LEDs, fixed flash pattern, you solder it yourself	Ready-built, 240 LEDs, real animations, warm white, EU seller with CE and GPSR accountability
Maker PCB ornaments (Tindie)	Christmas Tree PCB (ATtiny85 + WS2812)	**$3** bare PCB	Board only, build it yourself	Assembled and finished, far denser

Competitor type	Example	Price seen	What they offer	Our edge
Boutique PCB art (Tindie)	Christmas Tree Pre-built (coin cell, gold plating)	**$30**	Hand-made art piece, few LEDs	Same "PCB as art" appeal at under half the price, with far more motion
Mass-market LED decorations	Battery/USB acrylic or wooden LED trees in supermarkets and home-décor chains	typically €5–15 (indicative)	Static glow or simple flicker	Dense pixel animations, "tech" look, touch star

Takeaway: we cannot beat AliExpress on price. A $2.53 kit shipped is below our volume landed cost, and many such listings skip EU compliance costs that we must carry. We win on **looks, motion and being finished**, and we compete in the gift slot, not the bargain bin.

SWOT

Strengths	Weaknesses
Pine's personality: a Tamagotchi-like bond that grows (and is remembered) keeps it on the desk after Christmas and gets clips shared · 240-LED density nobody else has at this price · calm warm-white animations · cute rounded design · USB-C, no batteries · touch star · €0 customisation (silkscreen) · tiny BOM besides LEDs	Price can't match AliExpress · strongly seasonal · needs a cable/charger (not included) · some power banks switch off at ~20 mA · 1 k units don't carry EMC costs in retail
Opportunities	**Threats**
Pine can outlive the season as a year-round desk companion · B2B logo editions (September–October orders smooth the season) · Easter/Valentine/Halloween shapes on the same electronics (new outline = new product) · "hack it" firmware community · a later USB-data variant driven live from the simulator	Copycats: the design is easy to clone once seen · LED price spikes · GPSR enforcement also levels the field, since non-compliant marketplace sellers are being delisted

Go-to-market plan (first season)

1. **September:** finish routing, order 100-pc pilot, program, shoot videos of the animations. The simulator already renders them, so marketing clips can be made before hardware exists.
2. **Early October:** EMC pre-scan and full test (see eu-compliance/test-plan.md). Open B2B pre-orders with logo mock-ups.
3. **Mid-October:** 1 000-pc JLC Standard PCBA run (≈ 2 weeks production + 1 week air freight).
4. **November–December:** own e-shop + Etsy/Tindie, 2–3 Czech design/gift shops on consignment, Christmas-market stall.

Realistic first-season volume: 600–1 500 units (≈ 300 retail/e-shop + 1–2 B2B orders of 500). **Break-even on the €2 700 compliance budget** (contribution = revenue − €3.20 landed cost at 1 k): ~390 units through the own e-shop (€7.04 each), or ~730 units in B2B (€3.70 each).

Pricing recommendation

- **€12.90** shelf price (retail), **€14.90** own e-shop (covers shipping subsidy), **€11.90** "two-pack" per unit.
- **B2B:** €6.90 ex-VAT at 500 pcs, €5.90 at 2 000 pcs, logo included. Stand-fin and back-silkscreen changes are free; a custom outline adds a one-off €150 engineering fee.
- Don't discount below €9.90 retail. The product's appeal is "special", and a low price signals "AliExpress".

Pine: character, animations and firmware porting notes

The tree is a character called **Pine**. His job is to show you light shows. He's kinda bad at it: he loses focus, wanders off into his own thoughts (then only his face is on the tree), gets sleepy, and doesn't always want to be bothered. Touching the star reminds him to get back to work. How he reacts depends on how he feels and on how you've treated him, so your relationship with him changes over time.

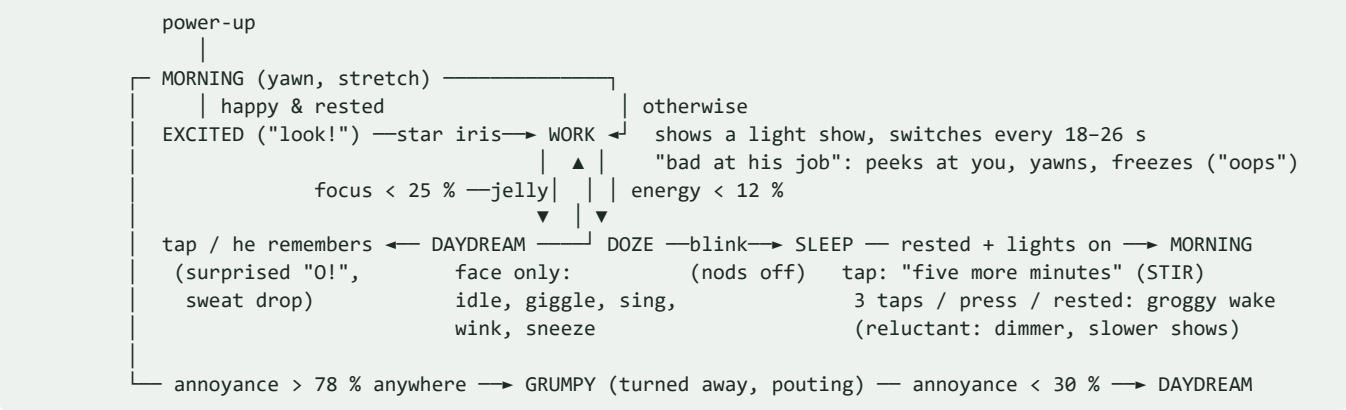
The reference implementation of everything below is `simulator/index.html` (sections *Pine's face*, *Face behaviours*, *Transitions*, *Touch star*, *Pine's brain*). Everything runs at the fixed firmware frame rate of **50 fps**, and brightness values are perceived brightness 0…1, gamma-corrected at the end.

Emotions (6 variables, 0…1, fixed-point in firmware)

Variable	Rises when	Falls when	What it changes
Energy	sleeping (faster in the dark, 2× while petted)	working, daydreaming (2× faster when the room is dark)	tired Pine yawns on the job, shows get dimmer and slower, below 12 % he dozes off
Focus	daydreaming refreshes him; a tap tops it up	working (slower when the bond is strong)	below 25 % he drifts away from the show into a daydream
Mood	petting, hugs, tickles (in moderation)	annoyance; being lonely for a long time	which daydreams he picks (giggle/sing when happy), cat mouth ω vs flat mouth
Annoyance	being woken, poked while grumpy, too much tickling	slowly by itself, fast while petted	above 78 % he turns away grumpy and ignores taps; below 30 % he forgives
Excitement	hover, hugs, tickles	decays quickly	— (feeds the "look what I made!" moment)
Bond	every gentle interaction; kept **in flash across power cycles**	very slowly if ignored for > 10 min	mood baseline, how often he peeks at you and winks, how quickly he loses focus

Rates at "1× like the real tree": energy lasts ~30 min of work, a nap takes ~3 min, focus drifts every few minutes. The simulator's "Pine's clock" speeds up only the feelings (10× by default). Show lengths, daydreams and reactions always run in real time.

Behaviour state machine



Reactions (short face scenes, then he goes back to what he was doing): hug (squeeze), squirm (squeezed while grumpy), tickled (giggle fit), petted (content ^ ^ ω, lasts while you hold), surprised.

Touch gestures (one star electrode, no extra parts)

The star is a single copper electrode **on both faces of the board, joined by a plated hole**, read by the ADC on PA7 (charge-sharing capacitance measurement). A finger on the front gives a medium signal. A pinch front + back roughly doubles the electrode area, which gives the largest signal, a "squeeze". The firmware classifier (identical to `touchTick()` in the simulator) turns the continuous signal into gestures:

Gesture	Signal (normalised Δcapacitance)	Timing	Pine's reaction (depends on state and feelings)
Hover	0.08–0.35 (finger ≤ ~2 cm away)	≥ 0.3 s	looks at you and smiles; peeks out of a show if the bond is good; "you're back!" after a long absence
Tap	0.35–0.65	< 0.45 s	working: next show · daydreaming: "Oh! Right, my job!" · asleep: stirs ("five more minutes"), wakes after 3 taps or if rested
Pet (light hold)	0.35–0.65	> 0.6 s	content face; mood up, annoyance melts, bond grows; deeper sleep
Press (firm)	0.65–1.0	> 0.15 s	insists: wakes him groggy (annoyed if he was really tired); "HEY!" when grumpy
Squeeze (pinch front + back)	> 1.0	> 0.08 s	hug: blush, sparkles, big mood + bond boost; squirms if grumpy
Tickle	3 taps within 1.4 s	—	giggle fit; the third tickle session in 25 s makes him annoyed ("st-stop it!")

The same thresholds are drawn on the simulator's sensor bar. The gesture buttons play recorded signals through the classifier; on the canvas, click = tap, hold = pet, shift+hold = press, ctrl+hold = squeeze.

Ambient light (for "room lights off, Pine gets sleepy") was planned as the LED-as-photodiode trick. Bring-up on the first board (2026-11-03) showed it **can't work in a full charlieplex**: every pair of lines carries two LEDs, one each way, so reverse-biasing one forward-biases its twin, and unbiased their photocurrents cancel. Rev A keeps the simulator's default "lights on"; rev B can free one LED pair by leaving one LED unpopulated (firmware README).

On the real board (same date) the star senses tap, pet, press, squeeze and tickle; hover (a finger ~1 cm away) is below the noise. Calibration and the measured readings are in the firmware README.

Face behaviours (Pine's faces)

Faces are drawn from small pixel parts on the LED grid: eyes at rows 14–15 (±3–4 columns), mouth at rows 17–18, blush at ±5–6. Parts: 9 eye shapes (open, wide, happy ^, sleep v, closed, heavy lid, squint > <, dot, sparkle), 10 mouths (smile, small, grin, cat ω, o, yawn, flat, pout, wavy, none), plus glyphs (z, ♪, sparkle, sweat drop, puff, "!"). The face can shift (nod, bounce, turn away) and the eyes can glance.

Face	Used for	Length
Happy Pine (idle)	daydream: blinks every 2–5 s, glances, blushes now and then; looks at you on hover	loop
Giggle	daydream when happy: ^ ^, grin, bounce, sparkles	loop
Singing	daydream when happy and awake: o/‿ mouth to a beat, notes float up, sways	loop
Wink	daydream, more often with a strong bond: wink + sparkle at the tree top	2.8 s
Snowflake sneeze	rare daydream: a flake drifts down onto his nose, > < … ACHOO, sparkle burst	7.5 s
Sleepy zzz	asleep: breathing glow, z's rising, occasional dream smile	loop
Five more minutes	tapped while asleep: one eye opens, grumble, rolls over	2.4 s
Dozing off	tired: heavy lids, nod, jolts awake, drifts off again	8 s
Grumpy	turned away, pouting, "hmph" puffs, squints when poked	loop
Excited	sparkly eyes, bouncing, "!": about to show you his favourite show	2.6 s
Oops!	caught daydreaming: wide eyes, "O", sweat drop	1.4 s
Good morning	power-up / waking: sleepy eyes → yawn → smile → stretch sparkle up to the star	6.2 s
Being petted	content ^ ^ ω, glowing cheeks	while held
Squeezed! / Squirm / Tickled	reactions to hug and tickle	1.5–3 s
Peek / Yawn	**overlays on a running light show** (the show dims while his face pops in)	2.2 / 1.8 s

Light shows (15)

Twinkle, Snowfall, Breathe, Spiral garland, Candle flicker, Star ripple, Fireflies, Rising sparks, Fill & drain, Northern lights, Fireworks, Ornaments, Soft radar, Heartbeat, Tinsel shimmer. Game of Life, heart eyes and **Scrolling text were removed** (nobody would be saying "Merry Christmas" now that Pine is a character). One show is Pine's **favourite** (random per power-up); he opens with it when he's excited to show you something.

Transitions

Transition	When Pine uses it
Star iris: a puffy star grows from the top and reveals the next scene	excited → showing you his favourite; waking up happy
Sleepy blink: eyelids close over the tree, then open on the next scene	dozing off, being woken reluctantly
Sparkle dissolve: pixels hop over one by one with a twinkle	"Oh right, my job!", next show on tap
Snow blanket: snow piles up over the old scene, melts over the new one	random show-to-show switch
Jelly wobble: rows sway sideways during the crossfade	his mind wandering off into a daydream
Soft crossfade	reactions, calm switches
Face morph	face → face changes don't wipe; each LED eases to the new expression, so eyes and mouth morph softly

Pine never uses the same random transition twice in a row.

Firmware budget: measured by a test compile

The product firmware now exists and runs on the board: `firmware/pixelpine/` (59 KB flash, 6.9 KB RAM, per-show CPU time measured on the MCU). It is a closer port than the sizing prototype below (double-precision brain, bit-exact `Math.sin`, the simulator's random streams), which is why it is larger than this estimate. The prototype's numbers are kept as the record of the MCU choice.

A bare-metal C port of everything that has to run on the tree lives in `firmware/sizetest/`: the scan ISR, 15 shows, 18 face behaviours, transitions, brain, touch classifier, ambient sensing and flash storage. The shared part (shows, faces, scenes, brain, tables) is identical for both MCUs. Only the drivers differ (startup, scan ISR, touch ADC, flash storage, linker script). It's a **sizing prototype, not tested on hardware**, but it's feature-complete enough that its size is real.

Result (`-Os`, whole-program link-time optimisation):

	CH32V003 (original)	PY32F003F18P6 (chosen)
Compiler	xPack GCC 15.2, RV32EC	Arm GNU Toolchain 14.2, Cortex-M0+ Thumb
Flash	20 856 B of 16 320 B: **128 %, doesn't fit**	**17 204 B of 64 512 B: 27 %**
RAM: static data + worst-case stack (call graph)	1 776 B of 2 048 B: 87 %	1 892 B of 8 192 B: 23 %
Soft-float code linked in	none	none

Where the flash goes (per module, compiled separately before LTO):

Module	CH32V003	PY32F003
Light shows	7 634 B	4 438 B
Faces	3 890 B	2 650 B

Module	CH32V003	PY32F003
Brain	3 252 B	2 424 B
Per-LED tables (incl. the 480 B anode-phase map)	2 717 B	2 717 B
Scenes + transitions	2 082 B	1 338 B
Drivers (startup, scan, touch, storage)	2 432 B	2 708 B

The shared code is 30–40 % smaller as Thumb code, mostly because the Cortex-M0+ multiplies in one instruction where RV32EC calls `__mulsi3`. The PY32 drivers are a little larger: the scan ISR uses a 512-byte lookup table to stay fast at 24 MHz, and flash is written a whole 128-byte page at a time from a small routine in RAM.

RAM fits on both because the outgoing scene of a transition is frozen into one 240-byte snapshot instead of running two live shows, and `build_masks()` works one scan phase at a time (16 B of stack instead of 256 B).

Options that were considered

Option	Flash	Cost per unit	Trade-off
A. Keep CH32V003, cut ~4 KB	16 KB, would end ~99 % full	$0.139 (4k)	Faces rewritten as keyframe tables (≈ −1.5 KB), 3–4 light shows dropped (≈ −2 KB), brain compacted (≈ −0.5 KB). No room left for new content.
B. Puya PY32F003F18P6 (Cortex-M0+, LCSC C55338319, 5 400 in stock)	**64 KB, 8 KB RAM; measured 27 % / 23 %**	**$0.128 (4k)**	Hardware multiply. No PLL, so without a crystal it runs at 24 MHz from the internal oscillator (32 MHz needs an external crystal), which is enough. Needs the pin map redone, which is cheap because the schematic and PCB are generated. 2-wire SWD instead of 1-wire: the SWD pins double as LED lines after boot, and the tree is programmed via connect-under-reset (5 pogo pads instead of 3).
C. Puya PY32F002AF15P6 (LCSC C54919781, 61 k in stock)	20 KB, 3 KB RAM; the measured build would use **84 % / 63 %**	$0.087 (4k)	Same core, same 24 MHz, so the PY32 drivers port almost unchanged, but it has a different TSSOP-20 pinout. Saves $0.041 per tree, but leaves only ~3 KB for new content.
~~WCH CH32V004 / V006~~	32 / 62 KB	—	Not stocked at LCSC; the V006 is a $3.46 pre-order at JLCPCB with no stock. Not viable today.

Decision (2026-09-24): B, the PY32F003F18P6. It's cheaper than the CH32V003 at volume ($0.128 vs $0.139 per 4 k reel), and the Arm test compile confirmed the fit: 17.2 KB of 64 KB flash and 1.9 KB of 8 KB RAM, which leaves ~46 KB for new shows and seasonal content. The schematic and PCB now use it: see the pin map in `hardware/scripts/geometry.py`. Option C was rejected: the F003's spare ~46 KB leaves room for firmware updates (new shows, seasonal packs) that the F002A couldn't take.

CPU time on the PY32F003 at 24 MHz (Cortex-M0+ with a single-cycle 32×32 multiplier, no divide, no FPU):

Load	Share of 24 MHz (estimate)
Scan ISR: 16 phases × 7 interrupts × 122 Hz ≈ 13.7 k/s × ~120 cycles (27 instructions + a 38-instruction `drive()` + ≈ 28 cycles M0+ entry/exit, counted from the disassembly), plus one 2 µs delay per phase	~ 8 %
`build_masks()` + gamma, 50 fps	~ 2 %
Heaviest show (Northern lights: 3 table-based noise samples per LED), 50 fps	~ 13 %
During a transition (outgoing scene frozen, no double rendering)	~ 13 %
Brain, faces, touch, ambient light	< 1 %

That's under 25 % at worst. At 24 MHz a 2 µs BCM slot is only 48 cycles, less than one interrupt. So the ISR shows bit 0 with a short counted delay and then bit 1 in the same interrupt, and only bits 2–7 get their own timer interrupt. Refresh stays at 122 Hz.

For comparison, the CH32V003 figures (RV32EC, no hardware multiply) from before the switch:

CPU time on the CH32V003 (RV32EC has no hardware multiply, divide or FPU, so every `*` is a ~30–40-cycle library call; the test build links `__mulsi3`, `__divsi3`, `__udivsi3`, `__modsi3` and `__umodsi3`):

Load	Share of 48 MHz (estimate)
Scan ISR: 16 phases × 8 bits × 122 Hz ≈ 15.6 k interrupts × ~40 cycles	~ 1.3 %
`build_masks()` + gamma, 50 fps	~ 1 %
Heaviest show (Northern lights: 3 table-based noise samples per LED), 50 fps	~ 11 %
During a transition (the outgoing scene is frozen, so no double rendering)	~ 11 %
Brain, faces, touch, ambient light	< 0.5 %

The simulator's `hash2()` uses two 32-bit multiplies per call; ported literally, Northern lights alone would use ~45 % of the CPU. The test build therefore uses a 256-byte permutation table and 8-bit lerps.

Porting rules: Q8.8 fixed point; sin/cos and exp(−x^2) from LUTs; per-LED geometry from const tables generated by `gen_tables.py` (plus cheap run-time derivations); value noise from a permutation table (no multiplies); constant divisors only (reciprocal multiply, preferably by powers of two); gamma via a 256-entry LUT (or x^2, $\gamma = 2$, to save 256 B); scan ISR per BCM slot with a pin → port-mode lookup table (≈ 120 cycles on the PY32, interrupt entry and exit included); optional current-sag compensation in `build_masks()`; 8-bit BCM with a 2 µs slot gives a 122 Hz refresh. Random numbers come from xorshift32 (shifts and XORs only) seeded by the ADC noise at power-up, so each tree has its own personality quirks.

Power & connectivity — PixelPine rev A

Power input

Item	Value
Connector	USB-C receptacle, 6-pin power-only (VBUS, GND, CC1, CC2), SHOU HAN TYPE-C 6P, SMD
Negotiation	2 × 5.1 kΩ pull-downs on CC1/CC2 (Rd), so any USB-C source treats the tree as a sink and turns VBUS on. Works with C-to-C chargers, laptop ports and USB-A-to-C cables. No USB-PD, no data lines.
Supply voltage	5 V nominal (USB: 4.75–5.25 V). The PY32F003 runs directly from VBUS (rated 1.7–5.5 V), **no regulator**
Protection	None beyond the MCU's own pin diodes and R19 on the touch pad. No fuse, TVS or reverse protection (USB-C can't be reversed electrically). A deliberate cost-down.
Cable / charger	Not included (EU common-charger logic: every household already has USB-C). Optional €0.30 cable for gift packs.
On/off	No switch. The tree runs whenever plugged in; Pine naps on his own (and more in a dark room).

How the LED current works

The 240 LEDs are charlieplexed on 16 GPIO lines with one 100 Ω resistor per line. The firmware scans **by anode**: in phase *p* line *p* is driven high and the cathodes of the LEDs to be lit are driven low, 16 phases per frame. When *k* LEDs share one phase slot, they share the anode pin's current:

$$I_{LED}(k) = (V - V_f) / ((R + R_{drv}) \cdot (k + 1))$$ $V = 5.0$ V, $V_f \approx 2.75$ V, $R = 100$ Ω, $R_{drv} \approx 30$ Ω

LEDs lit in the slot (k)	Per LED	Anode pin total
1	8.7 mA	8.7 mA
4	3.5 mA	13.8 mA
8	1.9 mA	15.4 mA
15 (all)	1.1 mA	**16.2 mA**, which is under the 20 mA PY32F003 pin rating

Only one phase is active at a time, so **LED supply current can never exceed ≈ 16.2 mA**, whatever the animation. The simulator's "Raw scan" mode shows this sag, and its "Compensated" mode shows the firmware correction.

Which LEDs share a phase

Which anode/cathode pair sits at which position is no longer a fixed interleaved pattern: `hardware/scripts/optimize_leds.py` picks it so the board routes on two layers (see the README). The price is that neighbouring LEDs sometimes share an anode line: 17 % of the LED pairs closer than 3.2 mm do (6 % with the old `anode = n mod 16`). So when a compact blob is lit, the busiest phase carries more LEDs and each of them gets less current. Average over random blob positions, with the per-LED current from the formula above:

Lit disc, radius	LEDs lit	Busiest phase k, old → now	Current per LED in it, old → now
3 mm	8	1.8 → 2.7	6.3 → 4.7 mA
5 mm	18	2.7 → 4.1	4.7 → 3.4 mA
8 mm	35	3.7 → 6.0	3.7 → 2.5 mA
12 mm	68	6.0 → 8.8	2.5 → 1.8 mA

The pin current limit is unaffected (still ≤ 16.2 mA, whatever the pattern). What changes is how unevenly a compact pattern's brightness is spread over the scan phases, which is what the "Compensated" mode is for: the simulator uses the real map, so its Raw/Compensated views already show the new behaviour.

Consumption

Per-animation figures come from the simulator's electrical model: 8-bit BCM, gamma 2.2, 100 % brightness, **20 s average** after a 5 s warm-up, sampled 10× per second. The MCU at 24 MHz is included as ≈ 3 mA (the datasheet gives 1.5 mA at 24 MHz with peripherals off; the rest is a margin for the timer, ADC and scan interrupts).

Animation	Average supply	Max supply	Average power @ 5 V	Peak pin current
Fireflies	3.4 mA	4.1 mA	17 mW	14.4 mA
Fireworks	3.5 mA	5.5 mA	18 mW	13.8 mA
Rising sparks	3.6 mA	4.0 mA	18 mW	15.9 mA
Snowfall	5.6 mA	7.1 mA	28 mW	15.4 mA
Star ripple	5.8 mA	6.8 mA	29 mW	15.1 mA
Ornaments	6.4 mA	7.7 mA	32 mW	15.1 mA
Heartbeat	6.9 mA	14.4 mA	34 mW	16.0 mA
Spiral garland	7.2 mA	8.8 mA	36 mW	15.6 mA
Candle flicker	8.6 mA	10.0 mA	43 mW	16.2 mA
Soft radar	8.7 mA	15.8 mA	44 mW	16.2 mA
Fill & drain	8.9 mA	12.8 mA	44 mW	16.2 mA
Tinsel shimmer	9.0 mA	11.9 mA	45 mW	16.2 mA
Breathe	10.6 mA	18.6 mA	53 mW	16.2 mA
Twinkle	11.4 mA	13.1 mA	57 mW	15.7 mA
Northern lights	11.8 mA	17.2 mA	59 mW	16.2 mA
Mean of all 15 shows	**7.4 mA**		**37 mW**	
Theoretical maximum (all 240 LEDs at 100 %)		**19.2 mA**	**96 mW**	16.2 mA
Pine asleep (sleep face + dim breathing glow, estimate)	≈ 4 mA		≈ 20 mW	—

Energy per season: 37 mW × 8 h/day × 45 days ≈ **13 Wh**, about €0.01 of electricity. The USB charger's own no-load loss is larger than the tree.

Light output: XL-1608WWC-06 is 780 mcd at 20 mA, about 2.4 lm per LED for a Lambertian emitter. With LED current capped at 16.2 mA, the whole tree emits **≈ 2 lm**. That's roughly one Christmas-light bulb spread over 240 pixels, and it keeps the product far below the 60 lm lower limit of the EU light-source ecodesign rules (see eu-compliance).

Thermal: 0.1 W on a 40 cm^2 board, so no measurable temperature rise.

Known limitation: power banks

Many power banks switch off when the load stays below ~50–100 mA for 10–30 s. PixelPine draws 3–19 mA (≈ 7 mA on average), so on those banks it goes dark after a few seconds. The cheap fix (**not fitted in rev A**) is an N-MOSFET (AO3400, C20917, JLC basic, ≈ $0.02) with a 47 Ω resistor. It would pulse 100 mA for 100 ms every 8 s. The only free pin is NRST (PF2), so this variant would have to program via the UART bootloader instead of connect-under-reset. Average cost: +1.25 mA, +6 mW. Recommended only if customer feedback asks for it; the leaflet says "use a USB charger or computer port".

User interaction: Pine and the touch star (no radio, no app)

The tree has a personality, **Pine**. His job is to show light shows, but he loses focus, daydreams, gets sleepy and has moods (full design in animations.md). You interact with him only through the star:

Gesture	How it's detected	Typical reaction
Hover (finger ≤ ~2 cm)	small capacitance rise, ≥ 0.3 s	Pine looks at you and smiles, peeks out of a show. *Rev A: below the noise, see below*
Tap	light touch < 0.45 s	next show · "Oh right, my job!" · a sleeping Pine stirs
Pet	light touch held > 0.6 s	content face, mood up, annoyance melts
Press	firm touch (flattened fingertip, bigger area)	insists: wakes him, next show, "HEY!" if grumpy
Squeeze	pinch front + back of the star (both faces are one electrode)	hug: blush and sparkles, or a squirm if grumpy
Tickle	3 taps within 1.4 s	giggle fit; too much makes him annoyed
Touch anywhere on the tree	*not on rev A*: the LED lines pick up a finger's mains hum only on a mains-tied supply (nothing on a powerbank), so it was dropped (see the firmware README)	—
Room lights off	*not on rev A*: the LEDs can't be used as light sensors in a full charlieplex (see the firmware README)	would make him sleepy in the dark
Power-up	—	"Good morning" yawn; remembers your **bond** (stored in flash)

Touch sensing. The star is a single electrode on both copper layers, joined by a plated hole, wired through R19 (1 kΩ, ESD) to **PA7, an ADC input**. The firmware measures its capacitance by charge sharing with the ADC's sample capacitor, which resolves fractions of a pF. That is enough to separate hover, touch, firm press and a two-sided pinch. A slow baseline tracker removes drift (humidity, temperature). Sampling happens between scan phases, so the LED matrix doesn't disturb it. No extra parts; the moved pin is the only hardware change.

Measured on the first board (2026-11-03): a resting finger drops the reading to ~44 % of untouched, a firm press to ~27 %, a pinch to ~20 %, so tap, pet, press, squeeze and tickle separate cleanly. A touching finger brings mains hum (the USB supply floats relative to the user), so the firmware takes ~39 samples spread over each 20 ms frame and averages them. Hover (a finger 1 cm away) is below the noise. Details in the firmware README.

Connectivity

Rev A has none, on purpose. No radio means:

- no Radio Equipment Directive (RED) and its cybersecurity delegated act, saving **€3–5 k** of extra testing,
- no USB-C data lines, so no USB compliance concerns and a cheaper 6-pin connector,
- no pairing, apps or firmware-update support burden.

Programming is done once in production through five bare pads on the back: 5 V, GND, SWDIO, SWCLK and NRST. A pogo-pin jig and any Arm SWD probe (CMSIS-DAP, J-Link, or a WCH-LinkE in Arm mode, ~$3) take ~5 s per board with *connect under reset*. While NRST is held low, PA13/PA14 are still in their SWD role. After boot the firmware takes them over as LED lines L8/L9, and PF4/BOOT0 as L15. Hackers can re-flash their own animations with the same pads.

Future variants (evaluated, not in rev A)

Variant	Extra parts	+ Cost	+ Compliance	What it enables
USB "Link"	16-pin USB-C, MCU → a USB-device MCU (e.g. CH32X035)	≈ $0.20	USB-IF optional; EMC unchanged	Firmware update by drag-and-drop, live control from the PC simulator
IR remote	IR receiver (needs one GPIO, so drop to 15 lines / 210 LEDs)	≈ $0.06 + remote $0.40	none	Change shows from the sofa

Variant	Extra parts	+ Cost	+ Compliance	What it enables
Music-reactive	MEMS mic + op-amp or PDM mic	≈ $0.15	none	Beat-synced animations
Bluetooth LE	CH582F / PY32 + BLE module instead of PY32F003	≈ $0.9	**RED + EN 300 328 + EN 18031 cybersecurity**, €3–5 k	Phone app, synchronised trees

Bill of materials — PixelPine rev A

All parts are stocked at LCSC and placeable by JLCPCB PCBA. Prices are LCSC list prices fetched **2026-09-23**, shown at the break a 1 000-board run reaches (incl. 3% assembler attrition on multi-placement parts). Machine-readable version for JLC: `hardware/production/jlc_bom.csv` + `jlc_cpl.csv`.

Ref	Part	LCSC	Qty	JLC class	Unit @1k	Per board
D1-D240	LED 0603 warm white 2500-3000 K, XINGLIGHT XL-1608WWC-06	C965810	240	extended	$0.0055	$1.36
U1	MCU Puya PY32F003F18P6 (TR), Cortex-M0+ (24 MHz on its internal oscillator), 64 KB flash, 8 KB SRAM, TSSOP-20	C55338319	1	extended	$0.148	$0.148
J1	USB-C receptacle 6-pin power-only, SHOU HAN TYPE-C 6P (SMD)	C456012	1	extended	$0.023	$0.023
C1	10 uF 10 V X5R 0603, Samsung CL10A106KP8NNNC	C19702	1	basic	$0.022	$0.022
R1-R16	100 R 1 % 0402 line resistors, UNI-ROYAL 0402WGF1000TCE	C25076	16	basic	$0.0008	$0.013
C2	100 nF 16 V X7R 0402, Samsung CL05B104KO5NNNC	C1525	1	basic	$0.0034	$0.0034
R17, R18	5.1 k 1 % 0402 USB-C CC pull-downs, UNI-ROYAL 0402WGF5101TCE	C25905	2	basic	$0.0018	$0.0037
R19	1 k 1 % 0402 ESD series R for the touch star (price assumed = C25076)	C11702	1	basic	$0.0010	$0.0010
	Total components		**263**			**$1.57**

Not purchased (bare copper, zero cost): TP1–TP3, TP5, TP6 programming pads (back), TP4 capacitive star (front).

Notes and substitutions

- **The LEDs are 86% of the component cost.** 240 × 0603 is the product; every other line is noise. A direct reel quote from XINGLIGHT or a comparable Shenzhen LED maker is the single biggest lever (≈ −30 %).
- **Cheaper, colder option:** KT-0603W (C2290) is a JLC *basic* white LED at $0.0060 @ 48 k — saves one extended-part fee ($3.07/order) but it is 40 000 K+ blue-white, not the warm look of the reference photo.
- **USB-C footprint:** `PixelPine:USB_C_SHOUHAN_TYPE-C-6P`, copied from LCSC's own EasyEDA footprint for C456012 with the same origin, so the CPL needs no offset. 6 SMD pads + 4 plated shell slots that are paste-in-hole reflowed with the rest of the board (JLC places it as an SMT part, no hand soldering).
- **No LDO, fuse, TVS or switch** — deliberate cost-down; the MCU is rated 1.7–5.5 V and runs from VBUS.
- **R19 (1 k)** was added after the EMC review: it protects the touch input (PA7) from ESD on the exposed star (≈ $0.004).
- **The star is a free sensor**: one copper electrode on both faces (a plated hole joins them) gives hover, tap, pet, press and a front-and-back squeeze; the LEDs double as an ambient light sensor. No sensor parts.

Cost analysis — PixelPine rev A

Planning rate **1 USD = 0.9 EUR**. Generated by `cost_model.py` — change an input there and re-run instead of editing numbers here. Items marked *EST* are engineering estimates; everything else is a list price (LCSC, JLCPCB published PCBA fees, fetched 2026-09-23).

Assembly volume per board: **552 SMD joints** (480 of them are LEDs), 0 THT joints, one side only.

Unit cost by volume

	Pilot · 100 pcs	Batch · 1 000 pcs	Volume · 10 000 pcs
How	JLCPCB Economic PCBA, parts from JLC/LCSC stock	JLCPCB Standard PCBA, panelised 2-up	Shenzhen contract manufacturer, LEDs bought direct
Components (BOM)	$1.75	$1.57	$1.14
Bare PCB incl. stand fin *(EST)*	$0.950	$0.400	$0.220
SMT assembly (joints + amortised setup)	$1.07	$0.717	$0.567
Programming (SWD pogo jig)	$0.0000	$0.090	$0.020
Stand fitting, test, packing labour	$0.0000	$0.190	$0.040
Packaging	$0.030	$0.180	$0.080
Ex-works cost	**$3.80**	**$3.15**	**$2.06**
Freight to CZ	$0.300	$0.240	$0.050
Landed in CZ (+3.7% duty, broker)	**$4.25 / €3.83**	**$3.55 / €3.20**	**$2.20 / €1.98**
Compliance amortised over the batch	$30.00	$3.00	$0.300
Total per unit	**$34.25 / €30.83**	**$6.55 / €5.90**	**$2.50 / €2.25**

Compliance budget (first year, one-off): EMC test at an accredited lab (EN IEC 55015 + EN 61547) €2,000, RoHS evidence (supplier declarations + 1 XRF screen) €400, WEEE + packaging EPR registration, first-year fees (CZ) €300 → **€2,700**. Details in eu-compliance/.

From cost to shelf price

Retail prices include 21 % Czech VAT. Retail channel assumes the usual keystone (retailer takes ~50 % of the net price); own e-shop assumes ~12 % payment/platform fees and a €0.60 shipping subsidy.

Channel	Shelf price	Net of VAT	Our revenue / unit	Margin @1k	Margin @10k
Retail / gift shop	€12.90	€10.66	€5.33	−€0.57	€3.08
Own e-shop, Etsy, Tindie	€14.90	€12.31	€10.24	€4.34	€7.99
B2B promo (custom silkscreen logo), 500+ MOQ	quoted ex-VAT	€6.90	€6.90	€1.00	€4.65

What the numbers say

- **The LEDs drive everything.** At 1 000 pcs the LEDs are 86% of the BOM, and their 480 joints are 87% of the assembly bill. Halving the LED count would cut the unit cost by roughly a third — but 240 dense LEDs *are* the product; keep them.

- **The MCU is almost free.** The PY32F003F18P6 (64 KB flash) is ~$0.13 in reels, cheaper than the 16 KB CH32V003 it replaced; a charlieplex driver chip such as the IS31FL3731 on the reference board costs over 10× more.
- **At 100 pcs compliance kills the margin** — treat it as a pilot (samples, school fair, beta testers) and keep those units off the market until the EMC report exists. At 1 000 pcs the classic retail channel loses money because the €2 700 compliance budget is spread over too few units; own e-shop and B2B promo orders carry it. From ~3 000 pcs retail works, and at 10 000 pcs every channel is healthy.
- **Competing with AliExpress** (bare soldering kits at $2.5–4.3 shipped) on price alone is not possible at the retail shelf — the landed volume cost is close to their selling price, and EU compliance plus VAT is a cost those listings often skip. The product has to win on looks (dense warm-white matrix, animations, ready-built) and sit in the €10–15 gift slot. See market-analysis.md.

Cost-down levers not yet taken

Lever	Saving / unit	Trade-off
Direct LED reel quote instead of LCSC list	≈ $0.40	needs 250 k+ LEDs per order
Cold-white basic LED (C2290)	≈ $0.05 + $3/order	loses the warm look
Drop C1 (10 µF) and rely on C2 + USB source capacitance	≈ $0.02	more supply ripple on touch sensing
Stand fin from the PCB panel's waste corners instead of its own area	≈ $0.03	tighter panel layout
Kraft paper sleeve instead of a printed box	≈ $0.10	less gift appeal

EU compliance pack — PixelPine rev A

Product: USB-C (5 V, ≤ 0.1 W) decorative LED Christmas tree, 240 LEDs, no radio, no battery, not a toy. Manufacturer placing it on the market: **MachStuff s.r.o.**, Jehličková 240, 240 24 Stromečkov, Czech Republic, marekmach24@gmail.com (the address is made up for the school project). That makes us the "manufacturer" under EU law even though assembly is subcontracted in China.

> This is an engineering-student compliance plan, not legal advice. Before the first sale, confirm the applicability matrix with the test lab or a notified-body consultant (a one-hour call, typically free with a test order).

Applicability matrix

Legislation	Applies?	Why	What we do
EMC Directive 2014/30/EU	**Yes → CE**	Electrical apparatus that can emit or suffer electromagnetic disturbance (fast GPIO edges, exposed touch pad)	Internal production control (Module A). Test to **EN IEC 55015** (emissions, lighting equipment) and **EN 61547** (immunity, lighting). Test plan
RoHS 2011/65/EU + (EU) 2015/863	**Yes → CE**	Electrical & electronic equipment	Lead-free HASL finish (**not** leaded HASL), RoHS declarations from LCSC for every part, 1 XRF screen of a finished unit. Technical file per **EN IEC 63000**
Low Voltage Directive 2014/35/EU	No	Only covers 50–1000 V AC / 75–1500 V DC; we run on 5 V DC	Electrical safety is handled under GPSR instead
General Product Safety Regulation (EU) 2023/988 (GPSR)	**Yes**	Consumer product; the LVD doesn't cover it	Risk assessment, technical documentation, product traceability markings, safety instructions in the local language, a contact for safety issues. Technical file
WEEE Directive 2012/19/EU (CZ: Act 542/2020 Sb.)	**Yes**	EEE, category 5 (small equipment)	Crossed-out-bin symbol **on the product** (back silkscreen ✓). Register as a producer and join a take-back scheme (e.g. ASEKOL, ELEKTROWIN, REMA) in every country we sell to
Packaging & Packaging Waste Regulation (EU) 2025/40 (PPWR)	**Yes, from 12 Aug 2026**	We place packaging on the market	Minimal, recyclable kraft box, no plastic window; EPR registration (CZ: EKO-KOM). Harmonised sorting labels come later (2028+)
REACH 1907/2006	**Yes**	Articles	Art. 33 SVHC check via supplier declarations; answer consumer SVHC requests within 45 days
Toy Safety Directive 2009/48/EC / Regulation (EU) 2025/2509	**No**	Annex I of both lists **Christmas decorations** as products *not* considered toys	Label "Not a toy. Decoration for indoor use." Design and marketing must not target children
Ecodesign for light sources (EU) 2019/2020	No	The regulation covers light sources of **60–82 000 lm**; the whole tree emits ≈ **2 lm** (calculation)	Keep the calculation in the technical file
Energy labelling (EU) 2019/2015	No	Same 60 lm scope as above	—
Radio Equipment Directive 2014/53/EU	No	No intentional radio	If a BLE variant ever ships: RED, EN 300 328, EN 18031 cybersecurity
Common charger (EU) 2022/2380	No	Amends RED, radio equipment only. We still use USB-C, in the spirit of it	—
Batteries Regulation (EU) 2023/1542	No	No battery	—
Product Liability Directive (EU) 2024/2853	Yes (liability)	Covers any product	Product-liability insurance (~€150–300/yr for a small seller); keep the technical file for 10 years

Legislation	Applies?	Why	What we do
Photobiological safety (EN 62471)	Hazard check only	Under GPSR, LED products are checked for blue-light / retinal hazard	≈ 2 lm of 2700 K light is trivially **Exempt Group**; document the reasoning, no test needed

Result: CE marking covers **EMC + RoHS** only. GPSR, WEEE, PPWR and REACH are "must do" obligations without their own mark (WEEE has the bin symbol).

Design decisions that were made *for* compliance

Decision	Regulation it serves	Cost
Lead-free HASL surface finish	RoHS	≈ +$0.02/board
R19 1 kΩ in series with the exposed star touch pad	EN 61547 ESD immunity (±4 kV contact / ±8 kV air)	$0.004
100 Ω on every charlieplex line + GPIO speed set to 2 MHz (firmware)	EN IEC 55015 radiated emissions: slow edges	€0, the resistors are needed anyway
No radio, no USB data	Avoids RED + cybersecurity testing	saves €3–5 k
Rounded board outline, no sharp tips	GPSR mechanical hazards (cuts, pokes)	€0, and it's cuter
Traceability marking on the back silkscreen (model, batch, manufacturer, address, e-mail)	GPSR Art. 9	€0
CE + WEEE logos on the back silkscreen	EMC/RoHS marking, WEEE Art. 14	€0

Roadmap and budget

#	Step	When	Cost (EUR)
1	Freeze rev A, route the board	T−8 weeks	—
2	Order 100-pc pilot; collect LCSC RoHS/REACH declarations for all 8 part numbers	T−7 w	in pilot cost
3	Write risk assessment + technical file draft (technical-file.md)	T−6 w	own time
4	**EMC pre-scan** at a university lab (near-field probe + spectrum analyser, ESD gun)	T−5 w	0–300
5	**Accredited EMC test** EN IEC 55015 + EN 61547 (e.g. EZÚ Praha, ITC Zlín, or a DE/AT lab)	T−4 w	1 500–2 500
6	XRF RoHS screen of one finished unit	T−4 w	~400
7	Sign the EU Declaration of Conformity (template)	T−3 w	—
8	WEEE producer registration + take-back scheme; EKO-KOM packaging contract	T−3 w	~300/yr + fees per kg
9	Print box and leaflet (texts)	T−3 w	in packaging cost
10	Product-liability insurance	T−2 w	150–300/yr
11	Start selling. For online shops: show the manufacturer's contact, product photo, warnings and identifiers on the listing (GPSR Art. 19)	T	—

Budget used in cost-analysis.md: **€2 700** (steps 5, 6, 8), plus insurance.

Files in this folder

- test-plan.md: EMC test list, setup, limits and risk points
- technical-file.md: technical documentation index + GPSR risk assessment
- declaration-of-conformity.md: EU DoC template, ready to fill in
- labelling-and-instructions.md: on-product marking, box texts, leaflet (EN + CZ)

EMC test plan — PixelPine rev A

Equipment under test (EUT): PixelPine rev A, standing on its stand fin, powered through a 1 m USB-A-to-C cable from a CE-marked 5 V USB adapter (reference AE, listed in the report), running the **worst-case EMC firmware mode**: all 240 LEDs cycling at 100 % duty, 8-bit BCM, fastest scan. This maximises edge rate and current steps. A second run uses the default auto playlist.

Standards for general lighting equipment are used because the product is a decorative luminaire. Our lab must agree this at quotation; the alternative multimedia-equipment pair EN 55032/55035 is **not** appropriate.

Emissions — EN IEC 55015:2019 + A11:2020

Test	Port	Range	Limit (class)	Relevance / risk
Radiated disturbance	Enclosure	30–300 MHz (CDNE or OATS/SAC at 10 m / 3 m)	Table 10/11 limits	**Main risk.** 16 GPIO lines toggle with µs-scale BCM slots; harmonics of the 2 µs slot clock (500 kHz) and GPIO edges reach the VHF band. Mitigation: GPIO speed 2 MHz, 100 Ω series R, short loops. Spread-spectrum dithering of the slot timer is a free firmware fallback.
Conducted disturbance	DC/USB power port (via the cable to the adapter)	9 kHz–30 MHz	Load/other-port limits per the lab's port classification	Low: ~10 mA average with 16 mA steps, and C1 10 µF buffers it
Conducted disturbance	AC mains	9 kHz–30 MHz	—	Measured on the **adapter**, which carries its own CE; state in the report that the EUT has no mains port
Harmonics / flicker (EN IEC 61000-3-2/-3)	AC mains	—	N/A	< 75 W and no mains port

Immunity — EN 61547:2009 (+ IEC 61000-4-x basic standards)

Test	Port	Level	Performance criterion	Relevance / risk
ESD IEC 61000-4-2	Enclosure: **touch star, front and back** (contact), board edges and USB shell (air)	±4 kV contact, ±8 kV air	**B** (temporary effect allowed, self-recovery)	**Main risk:** exposed copper (both faces) wired to ADC pin PA7. Mitigated by **R19 1 kΩ** + the MCU clamp diodes. Firmware: watchdog on, touch baseline re-learn after ESD glitches; a glitch may register as a gesture (Pine reacts), which is criterion B.
Radiated RF immunity IEC 61000-4-3	Enclosure	3 V/m, 80–1000 MHz, 80 % AM 1 kHz	A	Low: no analogue front-end; touch sensing may false-trigger, so debounce and require 3 consecutive samples
EFT/burst IEC 61000-4-4	DC power port	±0.5 kV (if the cable is > 3 m per the standard; ours is 1 m, so likely N/A)	B	Lab decides by port length
Conducted RF IEC 61000-4-6	DC power port	3 V, 0.15–80 MHz	A	Low
Surge IEC 61000-4-5	DC power port	N/A for USB-powered with cable < 30 m	—	—
Voltage dips	AC	N/A (no mains port)	—	—
Magnetic field IEC 61000-4-8	Enclosure	Only if magnetically sensitive parts, which we don't have	—	—

Performance criteria in plain language

- **A:** the animation keeps running without visible change.
- **B:** a flicker or a jump to another animation is allowed, but it must recover by itself with no settings lost.
- **C:** needs re-plugging. Not allowed for ESD in this plan.

Pre-compliance (before paying the lab)

1. **Near-field scan** of the trunk area and the charlieplex fan-out with an H-field probe and a spectrum analyser. Compare the full-on EMC firmware with a 25 % duty version.
2. **ESD gun** on the star: 10 discharges each at ±2, ±4 and ±6 kV contact. Pass = no reset or at most one glitch, and no latch-up (supply current returns to normal).
3. **Touch robustness:** 1 h next to a phone charger and a running laptop with no false touches.
4. Record everything in the technical file (photos of the setup, plots).

Samples and paperwork for the lab

- 3 finished units (one kept as a reference sample) + the USB adapter and cable used
- The EMC firmware build (hex) and a description of the operating modes
- Schematic PDF, BOM, photos, a draft of the user leaflet
- Quote request text: "*EMC testing of a 5 V USB-powered decorative LED luminaire (no mains port, no radio) to EN IEC 55015:2019+A11:2020 and EN 61547:2009, with test report suitable for CE Module A self-declaration.*"

Technical file — PixelPine rev A

Kept by the manufacturer for **10 years** after the last unit is placed on the market (EMC Art. 7, RoHS Art. 7, GPSR Art. 9). Market-surveillance authorities (CZ: ČOI, and ČTÚ for EMC) can request it in Czech or English.

Index

#	Document	Source / status
1	General product description, intended use, photos	this file (§ Description) + `hardware/renders/`
2	Schematic	`hardware/pixelpine_schematic.pdf` ✓
3	PCB layout, assembly drawings, BOM with manufacturer part numbers	`hardware/pixelpine.kicad_pcb`, `docs/bom.md`, `hardware/production/` ✓
4	List of standards applied	EN IEC 55015:2019+A11:2020, EN 61547:2009, EN IEC 63000:2018; EN 60598-2-20 and EN 62471 used as state-of-the-art references for the GPSR risk assessment
5	EMC test report	accredited lab, ◔ planned (test-plan.md)
6	EMC assessment (why the product meets the essential requirements, with design measures)	README.md § Design decisions ✓
7	RoHS: supplier declarations for all 8 LCSC parts + PCB (LF-HASL) + XRF screen	◔ collect from LCSC product pages / JLC
8	REACH SVHC statement	◔ supplier declarations
9	GPSR risk assessment	this file (§ Risk assessment) ✓
10	Photometric note: ≈ 2 lm, so outside 2019/2020 and 2019/2015	`docs/power-and-connectivity.md` ✓
11	EU Declaration of Conformity (signed)	declaration-of-conformity.md, ◔ sign after 5 and 7
12	Labelling, packaging and user information	labelling-and-instructions.md ✓
13	Production control: programming + functional test procedure, batch numbering scheme	this file (§ Production control) ✓
14	Firmware version identification	firmware version readable over SWD; version stored on the batch record

Description

A 70 mm tall, 1.6 mm FR-4 printed circuit board in the shape of a Christmas tree, with 240 warm-white 0603 LEDs, a PY32F003 microcontroller and a USB-C power-only receptacle. It stands on a slotted FR-4 stand fin. It is powered from any USB power source at 5 V and draws at most 19 mA (0.1 W). A capacitive touch electrode shaped as a star (on both faces of the board) lets the user interact with the tree's animated character. Indoor decorative use only. No battery, no radio, no mains connection.

Risk assessment (GPSR Art. 6 & 9)

Method: hazard → scenario → severity (1–4) × probability (1–5) → risk. Anything ≥ 8 must be reduced.

Hazard	Scenario	S	P	Risk	Measures	Residual
Electric shock	User touches the circuitry	1	1	1	SELV 5 V from a CE USB adapter; no mains inside	1
Overheating / fire	Short circuit from a metallic object on the board	2	2	4	Max 0.1 W normally. A USB source current-limits a short (USB 2.0: 500 mA+); FR-4 is UL94 V-0. Leaflet: "don't leave near flammable decorations while unattended" — conservative wording	2
Overheating	Faulty/cheap USB adapter delivering > 5.5 V	2	2	4	MCU abs. max 6.5 V; failure mode is a dead MCU, not fire (currents limited by 100 Ω line resistors). Leaflet: use a CE-marked adapter	2
Cuts / pokes	Sharp PCB tips	2	2	4	**Rounded outline** (min radius 0.6 mm at inner joins, 2.4 mm at tips), routed edges	1
Choking	Child swallows the					

stand fin or an LED knocked off | 3 | 2 | 6 | Not a toy, not marketed to children; warning on the box: "Not suitable for children under 3 years – small parts"; LEDs are reflow-soldered | 4 | | Eye (photobiological) | Staring at the LEDs from close up | 1 | 3 | 3 | ≈ 2 lm total, 2700 K, 1.1–8.7 mA per LED: Exempt Group per EN 62471 by inspection | 1 | | Photosensitive epilepsy | Flashing animation | 3 | 1 | 3 | No animation or face flashes above 3 Hz with large-area contrast (Fireworks bursts, sneeze sparkle and sparkle transitions are local or < 1 s; the fastest full-area change is a 0.9 s dissolve); design rule for all future content | 2 | | Tip-over | Tree knocked over on a table | 1 | 3 | 3 | Stand fin 36 mm; low mass; no hazard beyond the fall | 1 | | Chemical | Lead in solder / finish | 2 | 1 | 2 | RoHS: lead-free solder paste and LF-HASL | 1 | | Misuse outdoors | Used in rain | 2 | 2 | 4 | "Indoor use only" on product + box; 5 V cannot cause a shock | 2 |

Conclusion: no residual risk ≥ 8. The product is safe under normal and reasonably foreseeable use, with the listed warnings.

Production control (Module A — internal production control)

1. **Incoming:** JLC/CM supplies the certificate of conformity for the PCB (LF-HASL, UL94 V-0) and the reel labels (lot numbers) for the LEDs and MCU.
2. **Programming:** pogo jig on TP1–TP3, TP5, TP6 (SWD, connect-under-reset), flash, read-back verify.
3. **Functional test (8 s, automatic after flashing):** the test firmware lights each of the 16 scan phases in turn. The operator sees a "rolling" pattern, so a dead line shows as a missing diagonal. Then all 240 LEDs light dimly for a visual check of every LED. Then a touch test on the star.
4. **Batch numbering:** `YYWW-L` (year, week, line) written on the back silkscreen field with a paint marker, or laser-printed on the box label. The batch record stores firmware version, LED reel lot and test yield.
5. **Change control:** any change of the LED part, MCU, PCB outline, clock speed or scan timing triggers an EMC pre-scan and a review of whether the test report still covers the product.

EU Declaration of Conformity — template

Fill in the `<…>` fields, print on company letterhead, sign after the EMC report and RoHS evidence are in the technical file. Keep the signed original with the technical file. A copy (or a web link to it) accompanies the product when an authority asks. For RoHS/EMC it need not be in the box.

EU DECLARATION OF CONFORMITY · No. `<PP-DoC-001>`

1. **Product:** PixelPine, decorative LED Christmas tree
   Model / type: **PP-240-WW** · Hardware revision: **A** · Firmware: `<x.y>`
   Batch / serial range: `<from YYWW-L>`
2. **Name and address of the manufacturer:**
   MachStuff s.r.o., Jehličková 240, 240 24 Stromečkov, Czech Republic, IČO `<company ID>`, marekmach24@gmail.com
3. This declaration of conformity is issued under the sole responsibility of the manufacturer.
4. **Object of the declaration:** a USB-C powered (5 V DC, 0.1 W max) decorative PCB luminaire in the shape of a Christmas tree with 240 warm-white LEDs and a capacitive touch control, as shown in the photo `<attach photo>`.
5. The object of the declaration described above is in conformity with the relevant Union harmonisation legislation:
   - **Directive 2014/30/EU** (electromagnetic compatibility)
   - **Directive 2011/65/EU** (RoHS) as amended by Commission Delegated Directive **(EU) 2015/863**
6. **References to the relevant harmonised standards used:**
   - EN IEC 55015:2019 + A11:2020: limits and methods of measurement of radio disturbance characteristics of electrical lighting and similar equipment
   - EN 61547:2009: equipment for general lighting purposes, EMC immunity requirements
   - EN IEC 63000:2018: technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances
7. **Conformity assessment procedure:** internal production control (Module A, Annex II of Directive 2014/30/EU). No notified body involved. EMC test report: `<lab name>`, report no. `<…>`, dated `<…>`.
8. **Additional information:** the product is not a toy (Directive 2009/48/EC, Annex I). It is outside the scope of Regulations (EU) 2019/2020 and 2019/2015 (luminous flux ≈ 2 lm < 60 lm).

Signed for and on behalf of: **MachStuff s.r.o.**

Place and date of issue: Stromečkov, `<DD. MM. 2026>`

Name, function: `<Name Surname, jednatel / managing director>`

Signature: _____

Labelling, packaging and user information — PixelPine rev A

On the product (back silkscreen, already in the KiCad layout)

Marking	Required by	Status
CE logo (≥ 5 mm high)	EMC / RoHS	✓ 8.5 × 6 mm, KiCad `Symbol:CE-Logo_8.5x6mm_SilkScreen`
Crossed-out wheeled bin + bar	WEEE Art. 14, EN 50419	✓ 4.2 × 6 mm
Model **PP-240-WW**, rev A	EMC Art. 7(5), GPSR Art. 9(5)	✓
Batch / date field `Batch: ____ Date: __/__`	GPSR traceability	✓ filled by paint marker or replaced by a laser mark
Manufacturer name, postal address, e-mail	EMC Art. 7(6), GPSR Art. 9(6)	✓ MachStuff s.r.o., Jehličková 240, 240 24 Stromečkov, marekmach24@gmail.com (the address is made up for the school project: use the registered one for a real product)
Rating `5 V 0.1 A USB-C`	good practice	✓
"Indoor use only. Not a toy."	GPSR risk measures	✓

Box (70 × 100 × 25 mm kraft, one-colour print)

Front: PixelPine · *240 little lights. One tiny tree.* · product photo glowing

Back (EN / CZ):

> **PixelPine** — LED Christmas tree · Vánoční stromeček s LED
> Model PP-240-WW · 5 V ⎓ 0.1 A · USB-C (cable and charger not included · kabel a nabíječka nejsou součástí balení)
>
> ⚠ Not a toy. Not suitable for children under 3 years – small parts. Indoor use only.
> ⚠ Nejedná se o hračku. Nevhodné pro děti do 3 let – malé části. Pouze pro vnitřní použití.
>
> CE · crossed-out bin · packaging sorting symbols (paper) · MachStuff s.r.o., Jehličková 240, 240 24 Stromečkov, marekmach24@gmail.com · Made in China · Batch `YYWW-L`

Leaflet (A7, folded into the box)

EN

Hello, I'm Pine.

My job is to show you light shows. I'm... not always very good at it.

1. Slide the stand onto my trunk: the two slots lock into each other.
2. Plug a USB-C cable into my side and into any USB charger (5 V) or computer.
3. **Tap my star** when I drift off, to remind me of my job (or to see the next show).
4. **Hold it gently** to pet me. **Pinch it front and back** for a hug. I'm ticklish.
5. If I'm asleep, maybe let me sleep. And I remember how you treat me.

Safety

- Use only with a CE-marked USB power supply (5 V). Use indoors only; keep dry.
- Not a toy. Contains small parts; keep away from children under 3.

- Don't bend the board or place metal objects on it while it's powered.
- Some power banks switch off at very low loads; a wall charger or computer port works best.
- The LEDs are not replaceable. When I stop working, return me to an electronics collection point, not household waste.

Manufacturer: MachStuff s.r.o., Jehličková 240, 240 24 Stromečkov, marekmach24@gmail.com · Declaration of conformity: `<web link>`

CZ

Ahoj, já jsem Pine.

Mou prací je ukazovat vám světelné show. Moc mi to... nejde.

1. Nasuňte stojánek na můj kmen: oba zářezy do sebe zapadnou.
2. Zapojte USB-C kabel do mého boku a do libovolné USB nabíječky (5 V) nebo počítače.
3. **Klepněte na hvězdu**, když se zasním, ať si vzpomenu na svou práci (nebo pro další show).
4. **Jemně ji podržte** a pohladíte mě. **Stiskněte ji zepředu i zezadu** a obejmete mě. Mám lechtky.
5. Když spím, možná mě nechte spát. A pamatuju si, jak se ke mně chováte.

Bezpečnost

- Používejte pouze USB zdroj s označením CE (5 V). Pouze pro vnitřní použití, chraňte před vlhkostí.
- Nejedná se o hračku. Obsahuje malé části, uchovávejte mimo dosah dětí do 3 let.
- Desku neohýbejte a za provozu na ni nepokládejte kovové předměty.
- Některé powerbanky se při velmi malém odběru vypínají; nejlépe funguje síťová nabíječka nebo port počítače.
- LED nelze vyměnit. Po skončení životnosti odevzdejte výrobek na sběrné místo elektrozařízení, ne do komunálního odpadu.

Výrobce: MachStuff s.r.o., Jehličková 240, 240 24 Stromečkov, marekmach24@gmail.com · EU prohlášení o shodě: `<odkaz>`

Online listing (GPSR Art. 19, marketplaces and own e-shop)

Every listing shows: manufacturer name + postal & electronic address, a product photo, the model identifier PP-240-WW, and the warnings above in the language of the listing.

PixelPine product firmware

Runs Pine on the PY32F003F18P6: the 15 light shows, his face and 18 behaviours, the 6 transitions, the touch star, his brain and his bond, as the simulator (`simulator/index.html`) does. It's a port of the simulator's JavaScript, read side by side: the C names follow the JS ones. Written and brought up on the first rev A board on 2026-11-03.

Status: running on the board. The PC build of the same C code is checked frame by frame against the simulator's own JavaScript (see Matching the simulator). Touch is calibrated on one board and one person. Room-light sensing turned out to be impossible on rev A (see Room light), and a whole-tree touch field was tried and dropped (see Touch field).

Layout

Path	What
`src/shows.c`	the 15 light shows (`ANIMS[]`)
`src/face.c`	face parts, `renderPose()`, the 18 face behaviours (`FB`)
`src/scene.c`	scenes, overlays, the 6 transitions, `goTo()` / `engineTick()`
`src/touch.c`	the gesture classifier (`touchTick()`), unchanged thresholds
`src/brain.c`	emotions and the state machine (`brain`)
`src/pine.c`	one 50 fps frame: touch, brain, engine, gamma 2.2 to 8-bit levels
`src/fx.c`, `src/fx.h`	fixed-point maths: Q12 values, table sin / exp / pow, value noise, the simulator's hashes
`src/jsmath.c`	`Math.sin`, `Math.cos` and `%` bit-identical to V8 (fdlibm, the code V8 uses)
`src/tables.c`, `src/geom.h`	generated by `gen_tables.py` from `hardware/led_map.json`
`hw/`	PY32F003 drivers: startup, clocks, watchdog (`main.c`), BCM charlieplex scan (`scan.c`), touch (`sense.c`), bond in flash (`store.c`)
`host/host_main.c`	PC build for the tests
`test/`	`simref.js` (the simulator, headless), `compare.py` (the comparison), on-board tools

Build and flash

Needs the Arm GNU Toolchain (14.2), Python 3 and, for the tests, Node.js and MSYS2's mingw64 gcc. Flashing needs the J-Link setup in `../flash/`.

```
python gen_tables.py
```

```
python build.py flash
```

Command	Builds
`python build.py`	product firmware, `build/pixelpine.hex`
`python build.py flash`	the same, then programs the board through the J-Link
`python build.py host`	the PC test build, `build/pine_host.exe`
`python build.py showcycle`	bring-up: every show for 8 s in turn, no brain
`python build.py bench`	CPU-time bench; keeps SWD alive (lines L8/L9 dark), read with `test/readbench.py`
`python build.py diag`	LED test patterns, 6 s each: all off / top LED full / top LED dim / all at 64
`python build.py calib`	product firmware with SWD alive, for `test/sensemon.py`, `calibrate.py`, `gestures.py`, `stackcheck.py`

The product firmware turns SWDIO/SWCLK into LED lines L8/L9, so the debugger can only reach it through connect-under-reset (which `flash.py` does).

Resource	Used	Of
Flash	59 296 B (92 %)	63 KB (the last 1 KB holds Pine's bond)
RAM, static	6 896 B	8 KB
Stack, deepest measured	464 B (stack painting, `test/stackcheck.py`)	1 296 B left for it

How it maps onto the hardware

- **Scan:** 16 anode phases x 8 BCM bits, 2 us base slot: 16 x 255 x 2 us = 8.2 ms, **122 Hz** refresh (measured 119.6 Hz with interrupt overhead). This is the simulator's "Raw scan, 8 bit" model: each LED gets round(255 x brightness^2.2). TIM1 interrupts once per bit 2..7; bits 0 and 1 are shown from the phase-start interrupt with a counted delay. Masks are rebuilt every frame, one anode at a time with the interrupt held off.
- **Scan pitfalls found on the board (2026-11-03):**
  - **Pull resistors:** after reset PA13 (SWDIO) has a pull-up and PA14 (SWCLK) a pull-down. As lines L8/L9 they kept ~30 LEDs glowing faintly even with every LED at 0. `scan_init()` turns the pulls off on all 16 lines.
  - **Timer wrap:** if a bit-slot interrupt runs late, the new period can be shorter than the count already reached and the 16-bit timer runs to 65 535 (2.7 ms): bright flashes, worst in the radar. Guarded in the ISR.
  - **Latched phases:** each anode phase copies its 8 masks at its start, so a frame update mid-phase can't mix old and new bits.
  - **Empty slots:** a slot with no LED lit doesn't drive the anode (it would charge every floating line through the anode's LEDs, a faint flash on each).
  - **Current-sharing compensation** (the simulator's "Compensated" model) is in `scan.c` behind `-DSHARE_COMP` but off: on the board, and in the simulator's own model, it made levels jitter frame to frame.
  - **Debug builds never light LEDs:** in the `calib` / `bench` / `listen` builds SWDIO/SWCLK (lines L8/L9) stay on the probe, which holds SWDIO high, so every LED from L8 to a line the scan pulls low lit at full current (two "stuck LED" incidents). Those builds now keep every mask empty. All builds run the watchdog, and the crash handler floats every line first.
  - **Never halt a running board:** a halted core freezes the scan with one slot's pins driven, so one LED gets steady current (~8-9 mA, bright but within the LED's rating). Measure with connect-under-reset or the `calib` / `bench` builds without halting, and use `python build.py diag` for test patterns.
  - **Left over:** at 5 V a tiny current passes through two LEDs in series, so a lit LED can make one other LED glow very softly. Rev B fix: run the board a little below 5 V (a series diode, ~4.3 V).
- **Clock:** internal 24 MHz HSI (no PLL on the F003), 0 flash wait states. SysTick ticks the 50 fps frame.
- **Watchdog:** independent watchdog on the LSI (which has to be switched on first on the PY32), ~1 s, refreshed every frame; product build only, so the debug builds can halt. Checked with a test build (`-DKEEP_SWD -DWDT_TEST`): a deliberate hang from the debugger rebooted Pine after ~1 s.
- **Version:** `"PixelPine PP-240-WW fw 0.1.0"` at 0x080000C0, right after the vector table, readable over SWD.
- **Frame budget:** a frame is 20 ms. Most frames take 4-12 ms; a 205 s run of the normal firmware (calib build) had no late frame and a worst frame of 19.0 ms. A transition between the two heaviest shows with a face overlay can take up to ~24 ms; the main loop then runs the next frame's logic without rebuilding the LED levels, so animation timing stays locked to the simulator's frame count and the LEDs hold their levels for 20 ms. Measured on the board (bench build, 24 MHz):

Show	one frame alone		Show	one frame alone
Twinkle	2.3 ms		Fill & drain	4.1 ms
Snowfall	1.6 ms (+ flakes, ~5 ms full)		Northern lights	8.1 ms
Breathe	1.6 ms		Fireworks	3.8 ms
Spiral garland	4.3 ms		Ornaments	3.7 ms

Show	one frame alone		Show	one frame alone
Candle flicker	8.5 ms		Soft radar	2.9 ms
Star ripple	4.7 ms		Heartbeat	1.9 ms
Fireflies	1.3 ms		Tinsel shimmer	3.3 ms
Rising sparks	2.6 ms		Face overlay / face scene	~1-4 ms

- **Numbers:** per-LED loops run in fixed point (Q12, table sin / exp / pow). The brain, the face behaviours and the per-frame scalars stay in `double`, in the JS's order of operations, so their decisions are the same. Soft double on this core costs ~310 cycles per add or multiply, so particle positions (snow, sparks, fireworks, fireflies) moved to fixed point too.

Matching the simulator

`python test/compare.py` builds the PC version and runs it next to `test/simref.js`, which loads `simulator/index.html` into Node with the page stubbed out. Both get the same seeds, the same random-number stream (`Math.random` replaced by the firmware's xorshift32) and the same touch input (the simulator's recorded gesture signals). What is compared is what the LEDs get: the 8-bit level after gamma 2.2. Results on 2026-11-03:

`Math.sin`, `Math.cos` and `%`: **bit-identical** to V8 on 354 000 arguments.

Whole Pine from power-on (brain + faces + scenes + transitions + touch):

Scenario	Length	State machine	Emotions	LED levels within 1 step
Left alone, 1x clock	120 s	identical every frame	identical (diff 0)	100.00 %
Left alone, 10x clock	180 s	identical every frame (morning > excited > work > daydream > reminded > work > daydream > doze > sleep > morning > work)	identical	99.99 %
Touched a lot (hover, tap, pet, press, squeeze, tickle)	90 s	identical every frame	identical	99.99 %
Dark room, woken up, 30x clock	120 s	identical every frame	identical	100.00 %

Each light show alone, 30 s:

Show	exact	within 1 step		Show	exact	within 1 step
Twinkle	97.0 %	100.00 %		Fill & drain	96.8 %	99.37 %
Snowfall	99.0 %	99.99 %		Northern lights	80.7 %	99.43 %
Breathe	94.5 %	100.00 %		Fireworks	99.8 %	100.00 %
Spiral garland	95.3 %	99.93 %		Ornaments	99.8 %	100.00 %
Candle flicker	85.6 %	100.00 %		Soft radar	96.7 %	100.00 %
Star ripple	98.5 %	100.00 %		Heartbeat	93.8 %	100.00 %
Fireflies	89.0 %	93.31 %		Tinsel shimmer	93.0 %	98.62 %
Rising sparks	99.6 %	99.93 %				

Where it differs, and why:

- **Fireflies** follow the same steering rules, but their paths are chaotic: after ~5 s the fixed-point flies are somewhere else than the simulator's (the first 5 s match 97.5 %). The double-precision version matched to the end but cost 17 ms a frame.
- **Snowfall**: about once in 30 s a flake lands one frame earlier or on the neighbouring column (fixed-point flake positions), so that LED of the snow pile differs until it melts.

- **Northern lights, candle, tinsel, fill**: Q12 rounding in long chains of noise and powers; never more than 5 steps of 255.
- The simulator opens mid-yawn (it pre-runs 175 frames for the preview); the tree starts the yawn from the beginning.
- Pine's clock is the simulator's "1x like the real tree".

Touch calibration

Charge sharing between the star (PA7 through R19, 1 kOhm) and the ADC's sample capacitor. A touching finger also brings mains hum: the board's USB supply floats relative to you, and sampled at exactly 50 Hz the hum turns into a slow beat of several volts. So one sample is taken at every anode change of the scan (~39 per 20 ms frame) and the frame uses their mean. Two PY32 details: the ADC clears ADEN after every conversion (re-enabled per conversion), and ADEN is ignored for a few clocks after calibration.

Measured on the first board (`test/calibrate.py`, 2026-11-03, mean reading per frame, x8):

Gesture	Reading	Share of untouched
Untouched	8 510 +- 15	1.00
Finger resting on the front	~3 500-3 900	~0.44
Firm press, flattened fingertip	~2 200-2 500	~0.27
Pinch through front and back	~1 650-1 800	~0.20
Finger 1 cm above	no change	1.00

`hw/sense.c` maps the ratio piecewise-linearly onto the simulator's scale, with its thresholds halfway between these (touch 0.35 at a 18 % drop, press 0.65 at 0.35, squeeze 1.0 at 0.235). The touch threshold is low on purpose: contact takes a few hundred ms to build up, and a quick tap still has to count. The classifier is unchanged. A gesture test (`test/gestures.py`) recognised tap, tickle, pet and squeeze. A press has to be really firm, and a pinch passes through the press band on its way, so "press" fires just before "squeeze" (the simulator's recorded squeeze jumps straight to its level). A baseline tracker follows drift (~10 s), and something resting on the star for 30 s becomes the new baseline.

Hover can't be sensed with this electrode: a finger 1 cm away changes the reading by less than the noise. The gesture stays in the firmware (it fires if a finger comes very close).

Room light

The simulator's "room lights off, Pine gets sleepy" needs the tree to see the room light. The plan was to use the LEDs as photodiodes, but that **can't work in a full charlieplex matrix**: every pair of lines has two LEDs, one each way round. Reverse-biasing one LED forward-biases its partner, which clamps the line (that is what the first attempt measured: the same ~2.1 V with lights on and off), and without bias the two LEDs' photocurrents cancel. So rev A keeps the simulator's default, lights on: Pine still gets tired, dozes and sleeps on his own energy.

Fix for rev B, at no cost: leave one LED unpopulated (for example the L1 -> L0 one), so its partner L0 -> L1 has no anti-parallel twin and works as a photodiode between frames. The tree loses one pixel.

Touch field (tried, dropped)

The idea: use the 240-LED matrix itself as a touch surface, so a fingertip anywhere on the tree counts (Pine would look toward it, and a tap during a light show would send a ripple out from the finger). Tried on the board 2026-11-04 in five experiments and three runs of a full driver, then dropped: **the signal depends on how the tree is powered**, and on the supplies a customer would actually use it is too weak to be reliable.

Run	Method	Result
1	Lines as capacitive electrodes (charge sharing, RC time)	A fingertip changes a line's ~100-200 pF by <= 0.4 %: lost in the noise

Run	Method	Result
2	Listen for the mains hum a finger brings, every other line at 0 V	Strong and position-dependent: the finger moves the lines it sits on
3	9 spots; digital inputs vs ADC	Digital inputs never see it (the hum stays below the input threshold); the ADC separates 7 of 9 spots
4	Short listening gaps while Pine runs	Nothing: the lines still carry charge from the scan, which swamps the finger. 8 ms gaps flicker visibly
5	**Drain first**: every line to 0 V for 5 us, then float L0-L6 (analog mode, the lightest load) 20 us-2 ms	All 9 spots stand out, each with its own line pattern, rest reading flat (noise < 8 counts); 0.2 ms of float is enough. Board on a laptop's USB port
Driver	Drain gaps in the scan interrupt (2 per refresh, 0.23 ms, ~5 % of LED time), pattern match + interpolation	Powerbank: **no signal at all**. Wall charger: total rise 10-60 counts in one run, 7-14 in the next, against ~1 at rest; 3-4 of 9 spots placed right

Why it depends on the supply. The lines don't sense the finger's capacitance (run 1); they sense the 50 Hz hum a person carries, measured against the board's ground. On a laptop the board's ground is tied to mains and the difference is hundreds of counts. A USB wall charger links its output to mains only through a small capacitor, so the difference is weaker and changes with the charger and with how the person sits. On a powerbank the board floats along with the person and there is no difference to measure.

Why not keep it as an extra. At 7-14 counts a touch is only 3-7x the rest noise: a threshold low enough to catch it is close to false touches (a wrong rest level made the tree "touched" with no hand on it in the last run), and weak patterns are too flat to tell the spots apart. Nothing on rev A raises the signal much: a 2 ms float gives ~2x and costs visible LED time. A feature that works on some chargers and not others would read as a fault.

Found along the way:

- The PY32F003 has two comparators (COMP1 on PA1 = L1, COMP2 on PB6 = L11, against 1/4 Vrefint ~0.3 V). They do see the hum at a ~0.3 V threshold (experiment 5), but add nothing the ADC doesn't give.
- With the probe attached, SWDIO (line L8) is held high: any build that pulls lines low lights the L8 LEDs through the probe. The experiments ran with SWDIO / SWCLK unplugged and read their results once at the end.
- Polling the probe while the board runs pulsed RESET through the probe's wire and restarted the chip.

Rev B: a whole-tree touch would need dedicated electrodes (a copper pad or ring per zone) on free ADC pins, which the 20-pin PY32F003 doesn't have, or a self-capacitance touch IC. The touch star stays the control: it measures capacitance, so it works on any supply. The experiment code is in git history (commits `21a415c`, `b18460c`, `d70cac7`, `6f3f453`, `cee5f76`), and was removed from the tree afterwards.

Pine's bond

Saved to the last 1 KB of flash, an 8-page ring (128-byte pages; each save erases the next page and writes {magic, sequence, bond}). The simulator saves every 5 s; the tree saves when the bond moved by 0.02 or more, at most every 15 minutes, to spare the flash. Checked on the board: three saves rotated through pages 2-4 with rising sequence numbers, and after a reset Pine came back with the stored bond (0.392 instead of the default 0.3). The LEDs go dark for the ~4 ms page erase.

Tests

Command	Checks	Needs
`python test/compare.py`	firmware vs simulator, as above (~5 min)	Node.js, mingw64 gcc
`python test/readbench.py`	CPU time per show on the board	the `bench` build flashed
`python test/stackcheck.py`	deepest stack use	the `calib` build flashed
`python test/sensemon.py [s]`	live touch reading, Pine's state, late frames	the `calib` build flashed
`python test/calibrate.py`	guided touch calibration (spoken steps)	the `calib` build flashed

Command	Checks	Needs
`python test/gestures.py`	guided gesture recognition test	the `calib` build flashed

Not done

- Touch thresholds are from one board and one person; a production batch needs a spread measurement.
- The EMC worst-case firmware mode and the production test firmware (technical file) are separate builds still to write.

Firmware sizing prototype

> The product firmware is in `../pixelpine/`. This folder is the earlier sizing prototype that chose the MCU; it is kept as that record and isn't built for the board.

Purpose: answer "does everything fit in the MCU?" with a real compile instead of an estimate.

This is not the product firmware and it has **not been run on hardware**. The register-level code (scan ISR, ADC touch, flash writes) follows the vendors' reference headers and HAL sources but is untested.

It's a bare-metal C port of everything the tree has to run. The shared code is the same for every MCU:

File	What
`shows.c`	the 15 light shows, integer-only
`face.c`	Pine's face parts (bitmaps) and 18 behaviours
`scene.c`	scenes, face overlays, face morph and the 6 transitions
`brain.c`	emotions (Q16) and the behaviour state machine
`gen_tables.py` → `tables.c`, `geom.h`	per-LED constants from `hardware/led_map.json` (no float on the MCU), incl. which LEDs sit on which anode phase
`pp.h`, `libc.c`	shared declarations, `memset` / `memcpy`
`build.py`	builds each target, then reports flash, RAM, call-graph stack depth, soft-float check and the fit

The drivers are per MCU, in `py32f003/` (the product) and `ch32v003/` (the original choice, kept for the comparison):

File	PY32F003F18P6	CH32V003F4P6
`main.c`	vector table, HSI 24 MHz (no PLL), 50 fps SysTick loop, xorshift RNG	startup, 48 MHz PLL, 50 fps SysTick loop
`scan.c`	TIM1 charlieplex BCM scan ISR, bits 0+1 in one interrupt, 256-entry pin → `MODER` table	TIM2 scan ISR, one interrupt per bit
`touch.c`	ADC charge-sharing touch on PA7 (precharge from Vrefint), gestures, LED ambient-light sense	same on PD4
`store.c`	Pine's bond in a ring of 8 × 128-byte flash pages; page program from a RAM routine, as in Puya's HAL	16 slots in the last 64-byte page
`*.ld`	63 KB flash + 1 KB bond store, 8 KB RAM	16 KB − 64 B flash, 2 KB RAM

Build

```
python firmware/sizetest/build.py
```

It builds every target whose compiler it finds and skips the others:

- PY32F003: Arm GNU Toolchain (`arm-none-eabi-gcc`, tested with 14.2.Rel1 in `C:\Program Files (x86)\Arm GNU Toolchain arm-none-eabi\`)
- CH32V003: xPack RISC-V GCC (`riscv-none-elf-gcc`, tested with 15.2.0 in `%LOCALAPPDATA%\xpack\`)

Pass `py32f003` or `ch32v003` to build just one. Output goes to `build/<target>/`.

Result (2026-09-24)

	PY32F003F18P6 (product)	CH32V003F4P6 (original)
Flash, LTO	**17 204 B of 64 512 B (27 %)**	20 856 B of 16 320 B (128 %): doesn't fit
RAM: static + worst-case stack	**1 892 B of 8 192 B (23 %)**	1 776 B of 2 048 B (87 %)

	PY32F003F18P6 (product)	CH32V003F4P6 (original)
Soft-float code	none	none
Compiler warnings (`-Wall -Wextra`)	0	0

The CH32V003 overflowed, so the product MCU was changed to the PY32F003F18P6. The shared code is 30–40 % smaller as Cortex-M0+ Thumb code (single-instruction multiply instead of `__mulsi3` calls). The full breakdown, the CPU-time estimate at 24 MHz and the options that were considered are in docs/animations.md.

PixelPine — design spec (approved 2026-09-23)

A ~70 mm tall PCB Christmas tree covered in a dense charlieplexed matrix of 240 warm-white LEDs with software PWM, showing soft idle animations. Goal: lowest possible BOM + assembly cost ("compete with China"), price over reliability.

Hardware

Item	Decision
MCU	Puya PY32F003F18P6, TSSOP-20, LCSC C55338319. Cortex-M0+ at 24 MHz on the internal oscillator, runs straight from 5 V (no LDO). Was WCH CH32V003F4P6 until the firmware test compile, see the changelog.
LED matrix	16 GPIO charlieplex → 16×15 = **240 LEDs**. Pins: PA1, PA2, PC0–PC7, PD0, PD2–PD6.
LEDs	0603 warm white 2500–3000 K, XINGLIGHT XL-1608WWC-06, LCSC C965810 (JLC extended part, one feeder fee).
Current limit	16 × 100 Ω 0402 (C25076, basic), one per charlieplex line. Worst case common-pin current ≈ 15–17 mA (< 20 mA pin rating).
Touch	PD7 (NRST reconfigured as GPIO via option bytes) → bare-copper star pad at the tree top, RC-timing capacitive sensing. Cycles animations. Zero BOM cost.
Programming	PD1/SWIO + VDD + GND on 3 test pads (pogo jig, WCH-LinkE).
Power	USB-C 6-pin power-only receptacle, SHOU HAN TYPE-C 6P, LCSC C456012, 2 × 5.1 kΩ CC pull-downs (C25905). 10 µF 0603 bulk (C19702) + 100 nF 0402 (C1525) decoupling. No fuse, no TVS, no switch.
PCB	2-layer FR-4 1.6 mm, green mask, white silk, **lead-free HASL** (RoHS). Single-side SMT (top only).
Shape	3-tier fir silhouette, 70 mm total height incl. trunk; copper star on top; MCU/USB/resistors on the trunk. USB-C exits on the trunk's side edge.
Stand	Separate PCB "foot" fin in the same panel; cross-lap slot joint with the trunk (no solder, no parts).

LED drive scheme (for simulator + later firmware)

- Scan by anode: phase *p* (0–15) drives line *p* HIGH, lines of lit LEDs in that phase LOW, rest Hi-Z. 16 phases/frame.
- LED id = (anode a, cathode c), a≠c. The high pin sources all LEDs lit in its phase → brightness sags as k (LEDs lit simultaneously) grows: I_total(k) ≈ (V − Vf) / (R + R/k + Rdrv(1+1/k)), per-LED = I_total/k. Firmware/simulator can compensate.
- PWM: binary-code modulation, 6 bit default (64 levels) with gamma; target ≥ 200 Hz refresh.

Simulator (PC app)

Single self-contained `simulator/index.html` (no build, no server). Loads LED coordinates exported from the KiCad layout (`hardware/led_map.json` embedded as JS). Canvas render with glow, per-LED 0..1 brightness. Hardware-realism toggle: bit-depth quantisation, gamma, refresh rate, current-sag model. Controls: animation picker, speed, auto-playlist with crossfade, touch-star click = next animation. Animations are pure functions `(state, t, dt) → brightness[240]` using integer-friendly math so they port to C easily.

Approved animations

Twinkle, Snowfall, Breathe, Spiral garland, Candle flicker, Star ripple, Fireflies, Rising sparks, Fill & drain, Northern lights, Fireworks, Ornaments, **Soft radar** (wide, low-contrast glow sweep — softened per feedback), Heartbeat, Tinsel shimmer, Scrolling text, Auto-playlist. (Game of Life dropped.)

Deliverables

1. `hardware/` KiCad 10 project: schematic (ERC-checked), PCB with outline, stand, all footprints placed and routed (2 layers, Freerouting), renders.

2. `docs/BOM.csv` + `docs/bom.md` — LCSC/JLC part numbers, verified prices.
3. `simulator/index.html`.
4. `docs/market-analysis.md`.
5. `docs/eu-compliance/` — applicability analysis, test plan, DoC template, label/marking, technical file index.
6. `docs/cost-analysis.md` — 100 / 1k / 10k units incl. JLC assembly.
7. `docs/power-and-connectivity.md`.

Out of scope

Production firmware, real EMC testing.

Changes after approval (same day)

- **"Make it more cute"** (user request): puffy rounded star, chubby tiers with convex sides and rounded tips, rounded trunk and stand corners, silkscreen snowflakes; simulator restyled (Fredoka/Nunito, pill controls, background snow, wiggling star). LED grid refitted: 2.12 mm pitch.
- LED grid changed from hex to **square grid with LEDs at 45°**: at hex pitch the 0603 pads of 60° neighbours collide.
- **R19 1 kΩ** added between the star pad and PD7 for EN 61547 ESD immunity (~$0.004).
- USB-C C456012 uses `PixelPine:USB_C_SHOUHAN_TYPE-C-6P`, generated from LCSC's EasyEDA footprint (6 SMD pads + 4 plated shell slots, paste-in-hole). The 3D view borrows KiCad's GCT USB4125 model, which has the same body size.
- **Pine, the character** (user direction): the tree's job is showing light shows but he's bad at it, drifting into face-only daydreams, sleeping, getting grumpy or excited. Emotions: energy, focus, mood, annoyance, excitement, bond (bond persisted in flash). Faces: 16 behaviours; transitions: star iris, sleepy blink, sparkle dissolve, snow blanket, jelly wobble, crossfade, plus face morphing. Heart eyes and Scrolling text removed (15 light shows remain).
- **Richer touch**: touch moved from PD7 (RC timing) to **PD4 / ADC A7** (charge sharing, sub-pF resolution); PD7 becomes charlieplex line L13. The star footprint became a two-sided pad joined by a 0.6 mm plated hole, so a front+back pinch reads as "squeeze". Gestures: hover, tap, pet, press, squeeze, tickle. Ambient light from the LEDs.
- **Schematic**: blocks are now wired (MCU → line resistors, decoupling on VDD, USB-C → CC resistors / bulk cap, LED rows on an anode bus); net labels only cross between blocks.
- **MCU swapped to Puya PY32F003F18P6** (2026-09-24). A test compile of the full firmware measured 20.3 KB flash and 1.8 KB RAM, which overflows the CH32V003 (16 KB). The PY32F003F18P6 (LCSC C55338319) has 64 KB flash and 8 KB RAM, a Cortex-M0+ with a hardware multiplier (24 MHz: it has no PLL, so 32 MHz would need a crystal), and costs $0.128 at 4 k (vs $0.139). TSSOP-20 pinout 1: lines on PA0–PA6, PA12, PA13/SWDIO, PA14/SWCLK, PB5–PB7, PF0, PF1, PF4/BOOT0; touch on PA7 (ADC); NRST kept for connect-under-reset SWD programming (5 pads). Project symbol generated from the datasheet.
- **PY32F003 test compile** (2026-09-24): the same firmware built with the Arm GNU Toolchain 14.2 for the PY32F003 measures 16.7 KB flash (26 %) and 1.9 KB RAM (23 %). The size test now has drivers for both MCUs.
- **USB-C footprint** (2026-09-24): J1 now uses `PixelPine:USB_C_SHOUHAN_TYPE-C-6P`, generated from LCSC's EasyEDA footprint for C456012, replacing the GCT USB4125 placeholder.
- **Markings** (2026-09-24): manufacturer MachStuff s.r.o., Jehličková 240, 240 24 Stromečkov (made-up address) on the back silkscreen and in the EU documents; the stand fin now says "PixelPine" instead of "Merry Christmas" (Pine is the character, so nobody would be saying it).

PixelPine Implementation Plan

> Executed inline in the authoring session (user asked for minimal interruptions).

Goal: Produce every deliverable in the spec: KiCad schematic + placed PCB, BOM, browser simulator, market / cost / power / EU-compliance docs.

Architecture: One Python geometry module is the single source of truth for the tree outline, LED positions and the charlieplex map. Generators consume it to write the KiCad schematic (s-expression text) and the PCB (KiCad's bundled `pcbnew` Python API), and to export `led_map.json` for the simulator. The simulator is a single static HTML file.

Tech Stack: KiCad 10.0 (`kicad-cli`, bundled Python 3.11 + `pcbnew`), vanilla HTML/Canvas/JS, Markdown.

Spec: `docs/superpowers/specs/2026-09-23-pixelpine-design.md`

Global Constraints

- 240 LEDs = 16×15 charlieplex on PY32F003F18P6 PA0–PA6, PA12, PA13, PA14, PB5–PB7, PF0, PF1, PF4; touch on PA7 (ADC); NRST kept; SWD programming.
- LED id `n` ↔ (anode a, cathode c): a = n mod 16, c = (a + 1 + n div 16) mod 16 (interleaved phases). *Superseded:* the pair at each position now comes from `hardware/led_assignment.json`, chosen so the board routes on 2 layers (see the README).
- Total tree height 70 mm incl. trunk; top-side-only SMT; lead-free HASL.
- LCSC parts: C5187096, C965810, C25076, C25905, C456012, C19702, C1525.

Task 1: Geometry + LED map — `hardware/scripts/geometry.py`

- Produces `outline_polygon() -> list[(x_mm, y_mm)]`, `led_positions() -> list[(x, y, rot_deg)]` (exactly 240, sorted top→bottom, left→right), `charlie(n) -> (a, c)`, `foot_outline()`, `STAR_POLY`, and constants.
- Writes `hardware/led_map.json` `{ "leds": [{"id","x","y","a","c"}], "outline": [...], "star": [...] }`.
- Verify: script asserts 240 LEDs, all inside the outline with ≥0.9 mm clearance, and min spacing ≥ 2.2 mm.

Task 2: Schematic — `hardware/scripts/gen_sch.py` → `hardware/pixelpine.kicad_sch`

- Embeds lib symbols from KiCad stock libs (project PY32F003F18P6 symbol, LED, R, C, USB_C_Receptacle_PowerOnly_6P, TestPoint, power symbols).
- Two logical areas: MCU/power/programming, and the 16×15 LED matrix drawn as a grid with net labels L0..L15.
- Verify: `kicad-cli sch erc` → 0 errors; `kicad-cli sch export pdf` + svg render inspected.

Task 3: PCB — `hardware/scripts/gen_pcb.py` → `hardware/pixelpine.kicad_pcb`

- Board outline (tree + foot, same panel), slots, footprints placed with nets from the same map, star pad, test pads, silkscreen.
- Verify: `kicad-cli pcb render` / `export svg` images inspected; netlist consistency checked by script.

Task 4: Simulator — `simulator/index.html`

- Animation interface: `{ name, init(ctx) , frame(ctx, t, dt, out:Float32Array(240)) }`, ctx has `leds` (x,y normalized), `rand()`, helpers.
- Hardware model: BCM bit depth, gamma, current sag per anode phase, refresh.
- Verify in the built-in browser: every animation renders, no console errors, playlist + touch-star cycle works.

Task 5: Docs — BOM, cost, market, power, EU compliance

- BOM prices verified live on LCSC; cost model at 100 / 1k / 10k.

- Verify: numbers in cost doc reconcile with BOM (script-generated tables).

Ordering PixelPine rev A at JLCPCB

Generated 2026-10-02 from `hardware/pixelpine.kicad_pcb` **as saved**, hand edits included (GND pour on both layers, extra vias, rotated test pads). Two orders: the tree with assembly, and the stand fin as a bare PCB.

File	Use
`pixelpine_tree_gerbers.zip`	**Order 1**: the tree, PCB + assembly (Gerbers + Excellon drill)
`jlc_bom.csv`, `jlc_cpl.csv`	**Order 1**, assembly step: bill of materials and placement file
`pixelpine_stand_gerbers.zip`	**Order 2**: the stand fin, PCB only (no parts, no holes)
`preview_tree_top/bottom.png`, `preview_stand_top/bottom.png`	3D renders of exactly what went into the Gerbers
`gerbers_tree/`, `gerbers_stand/`	the same files unzipped, for a Gerber viewer

Regenerate after editing the board in KiCad (it doesn't change the board):

```
"/c/Program Files/KiCad/10.0/bin/python.exe" hardware/scripts/export_gerbers.py
```

```
python hardware/scripts/export_fab.py
```

Why two orders: the fin is a separate piece in the KiCad file, and JLC counts every piece that can be separated (milled, V-cut or tabbed) as an extra design, which costs extra (JLC: what counts as different designs). If you'd rather have one order, upload the tree zip with *Different design = 1* and order the fin separately later. Both pieces must be **1.6 mm** thick: the cross-lap slots are 1.7 mm wide.

Order 1: the tree (PCB + assembly)

Upload `pixelpine_tree_gerbers.zip` on jlcpcb.com → *Order now*. Settings (everything not listed: default):

Setting	Value	Why
Base material / Layers	FR-4 / **2**	
Dimensions	50.53 × 70.00 mm	read from the Gerber
Different design	1	
Delivery format	Single PCB	
PCB thickness	**1.6 mm**	the stand fin's slot is cut for 1.6 mm
PCB color / Silkscreen	Green / White	
Surface finish	**LeadFree HASL**	RoHS (compliance plan)
Outer copper weight	1 oz	
Via covering	Tented	
Min via hole size / diameter	0.3 mm / (0.4/0.45 mm)	the board uses 0.3 mm drills, 0.4 mm pads: no surcharge
Board outline tolerance	±0.2 mm (regular)	
Mark on PCB	**Remove Mark**	otherwise JLC prints its order number wherever it likes; the board has no marker for "specify position" (JLC ordering notes)
Electrical test	Flying probe	

Then switch on **PCB Assembly**:

Setting	Value
PCBA type	Economic
Assembly side	Top side

Setting	Value
Tooling holes	Added by JLCPCB (the board has none)
Confirm parts placement	**Yes**: check the preview below before production

Upload `jlc_bom.csv` and `jlc_cpl.csv`: 8 part numbers, 263 placements per board (240 LEDs, 19 resistors, 2 capacitors, USB-C, MCU). The test pads and the touch star are bare copper and aren't in the files.

In the placement preview, check:

- **LEDs D1–D240:** each LED's cathode mark must sit on its **lower-left** pad (pad 1, seen from the front). LCSC's footprint for C965810 has the cathode on pad 1, the same as KiCad's, so the 45° rotations in the CPL need no correction. Look at a few LEDs in every row.
- **U1 (PY32F003, TSSOP-20):** the pin-1 dot belongs at the **top-left** corner, next to the ▶ printed on the board. LCSC's footprint for this part is drawn turned 90° compared with KiCad's, which usually means rotation 270 instead of 0. It looked right in the preview on 2026-09-24, so the CPL keeps 0. If it shows turned, fix it in the preview.
- **J1 (USB-C):** the mouth faces the trunk's right edge. Its footprint is LCSC's own, so it lines up.
- Resistors and capacitors aren't polarised.

Stock: one board takes 240 LEDs (XL-1608WWC-06, C965810). Check LCSC has 240 × boards plus JLC's attrition before ordering. LED, MCU and USB-C are *extended* parts (a small fee per part type).

Order 2: the stand fin (PCB only)

Upload `pixelpine_stand_gerbers.zip`. Same settings as the tree, **no assembly**:

Setting	Value
Layers / Dimensions	2 / 36.00 × 10.00 mm (no copper, silkscreen only)
PCB thickness	**1.6 mm**
Surface finish	LeadFree HASL (there's no exposed copper; keeps the RoHS paperwork simple)
Mark on PCB	Remove Mark

Board check (KiCad DRC, 2026-10-02)

- 0 unconnected items, 0 schematic-parity issues, zone fill up to date.
- 2 *starved thermal* errors: J1 pad B12 and R18 pad 2 reach the GND pour through one thermal spoke instead of the required two. Both pads also have their own GND track, so the connection is solid. This is cosmetic, and JLC doesn't check for it. To clear it, give those two pads a solid zone connection in KiCad, or set the zone's minimum spoke count to 1.
- Design rules match JLC's standard 2-layer process (no surcharges): 0.10 mm minimum track and space, 0.3 mm drill / 0.4 mm vias, 0.3 mm copper-to-edge.

Don't run `hardware/scripts/build_all.py` on this board: it regenerates the PCB from the scripts and would discard the hand edits. It now refuses to run unless you pass `--regenerate`.